

T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Field combinations 1 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 \ D$	S_2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\begin{aligned} & -\frac{\overline{2hy_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1^\dagger\right)^2}{\epsilon} \\ & +C_{S_1^\dagger S_1}S_1S_1^\dagger \\ & -C_{S_1^\dagger S_1}\overline{hy_{1,i_1,r_1}S_1y_{1,i_1,r_1}S_1^\dagger} \\ & -\frac{2C_{S_1^\dagger S_1}\overline{hy_{1,i_1,r_1}S_1y_{1,i_1,r_1}S_1^\dagger}}{\epsilon} \\ & +\frac{1}{2}S_1^2\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)} \\ & -\frac{\overline{2hy_{1,i_1,r_1}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2}\lambda_{S_1}^{(1)}}{\epsilon} \\ & -\overline{hy_{1,i_1,r_1}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2}\lambda_{S_1}^{(1)} \end{aligned}$	+D_{\mu_1}(S_1)^\dagger D_{\mu_1}(S_1)	$\begin{aligned} & +\frac{\overline{2hy_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{3\epsilon} \\ & +C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_2}^\dagger \\ & -\frac{C_{S_2^\dagger S_2}\overline{hy_{2,i_1,r_1}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_2}^\dagger}}{\sqrt{3}} \\ & -\frac{2C_{S_2^\dagger S_2}\overline{hy_{2,i_1,r_1}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_2}^\dagger}}{\sqrt{3}\epsilon} \\ & -\frac{\overline{4hy_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon} \\ & +\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)} \\ & -\frac{\overline{2hy_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}\mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon} \\ & -\frac{\overline{hy_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}\mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}} \\ & +\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)} \\ & -\frac{\overline{2hy_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}\mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon} \\ & -\frac{\overline{hy_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}\mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}} \end{aligned}$
cont.										
cont.										
cont.										
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\begin{aligned} & -\frac{\overline{2hy_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1^\dagger\right)^2}{\epsilon} \\ & +C_{S_1^\dagger S_1}S_1S_1^\dagger \\ & -C_{S_1^\dagger S_1}\overline{hy_{1,i_1,r_1}S_1y_{1,i_1,r_1}S_1^\dagger} \\ & -\frac{2C_{S_1^\dagger S_1}\overline{hy_{1,i_1,r_1}S_1y_{1,i_1,r_1}S_1^\dagger}}{\epsilon} \\ & +\frac{1}{2}S_1^2\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)} \\ & -\frac{\overline{2hy_{1,i_1,r_1}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2}\lambda_{S_1}^{(1)}}{\epsilon} \\ & -\overline{hy_{1,i_1,r_1}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2}\lambda_{S_1}^{(1)} \end{aligned}$	+D_{\mu_1}(S_1)^\dagger D_{\mu_1}(S_1)	$\begin{aligned} & +\frac{\overline{2hy_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{3\epsilon} \\ & +C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_2}^\dagger \\ & -\frac{C_{S_2^\dagger S_2}\overline{hy_{2,i_1,r_1}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_2}^\dagger}}{\sqrt{3}} \\ & -\frac{2C_{S_2^\dagger S_2}\overline{hy_{2,i_1,r_1}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_2}^\dagger}}{\sqrt{3}\epsilon} \\ & -\frac{\overline{4hy_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon} \\ & +\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)} \\ & -\frac{\overline{2hy_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}\mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon} \\ & -\frac{\overline{hy_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}\mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}} \\ & +\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)} \\ & -\frac{\overline{2hy_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}\mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon} \\ & -\frac{\overline{hy_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}\mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}} \end{aligned}$
cont.										
cont.										
cont.										
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & -\frac{\overline{2hy_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1^\dagger\right)^2}{\epsilon} \\ & +C_{S_1^\dagger S_1}S_1S_1^\dagger \\ & -C_{S_1^\dagger S_1}\overline{hy_{1,i_1,r_1}S_1y_{1,i_1,r_1}S_1^\dagger} \end{aligned}$	+D_{\mu_1}(S_1)^\dagger D_{\mu_1}(S_1)	$\begin{aligned} & +\frac{\overline{2hy_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{3\epsilon} \\ & +C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_2}^\dagger \\ & -\frac{C_{S_2^\dagger S_2}\overline{hy_{2,i_1,r_1}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_2}^\dagger}}{\sqrt{3}} \end{aligned}$

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 \ D$	S_2	
cont.									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$		
									$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$		
T3-D	0	3	0	1	1	2	$\frac{1}{2}$		$+\frac{2\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$ $-\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{\epsilon}$ $+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$ $-C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$ $-\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$ $+\frac{1}{2}S_2^2\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}^2y_{2,i_1,r_1}\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}}{\epsilon}$
	cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$		
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$		$-\hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}^2y_{2,i_1,r_1}\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}$
	cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$		
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1		$+\frac{2\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$ $-\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{\epsilon}$ $+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$ $-C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$ $-\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$ $+\frac{1}{2}S_2^2\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}^2y_{2,i_1,r_1}\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}}{\epsilon}$
	cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$		
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$		$-\hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}^2y_{2,i_1,r_1}\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}$
	cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$		
T3-D	2	3	1	1	2	2	$\frac{3}{2}$		$+\frac{2\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$ $-\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{\epsilon}$ $+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$ $-C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
	cont.										

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
										$-\frac{2C_{S_1^\dagger S_1} \overline{h\bar{y}_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\bar{y}_{1,i_1,r_1} \overline{y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{2C_{S_2^\dagger S_2} \overline{h\bar{y}_{2,i_1,r_1}} S_{2,j_2} y_{2,i_1,r_1} S_{2,j_2}^\dagger}{\epsilon}$ $+\frac{1}{2} S_2^2 \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}$ $\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
cont.										$-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
cont.										$-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
cont.										$-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
										$+\frac{2h\bar{y}_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_1^\dagger S_1} S_{1,j_1} S_{1,j_1}^\dagger$ $-\frac{C_{S_1^\dagger S_1} \overline{h\bar{y}_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_1^\dagger S_1} \overline{h\bar{y}_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\bar{y}_{1,i_1,r_1} \overline{y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$		$+D_{\mu_1} \left(S_{1,j_1}\right)^\dagger D_{\mu_1} \left(S_{1,j_1}\right)$	$+\frac{2h\bar{y}_{2,i_1,r_1} y_{2,i_2,r_2} S_{2,j_2}^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_2^\dagger S_2} S_{2,j_1} S_{2,j_1}^\dagger$ $-\frac{C_{S_2^\dagger S_2} \overline{h\bar{y}_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_2^\dagger S_2} \overline{h\bar{y}_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\bar{y}_{2,i_1,r_1} \overline{y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^2\lambda_{S_2}^{(2)}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{4h\bar{y}_{2,i_1,r_1} \overline{y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\d$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
cont.									$-\frac{2h\overline{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} 2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{2h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} 2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$
									$-\frac{h\overline{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} 2\lambda_{S_1}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} 2\lambda_{S_2}^{(2)}}{\sqrt{3}}$

Field combinations 2 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	D	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$			$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{2\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{9\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$			
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{9\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$			$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{3\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{27\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$			$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{2\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{9\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	D	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0		$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$		$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1		$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{2\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$		$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{9\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} \epsilon M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger}{\epsilon}$	$-\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_2} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3} \epsilon}$
								$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$-2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_2,r_2} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} \epsilon M_F}$	$+H_{a_2} S_1 H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} \lambda_{H1}^{(1)}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
cont.								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} \lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
									$-2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} \lambda_{H1}^{(1)}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
									$-\frac{2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} \lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} \epsilon M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger}{\epsilon}$	$-\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_2} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3} \epsilon}$
								$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$-2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_2,r_2} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} \epsilon M_F}$	$+H_{a_2} S_1 H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} \lambda_{H1}^{(1)}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
cont.								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} \lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
									$-2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} \lambda_{H1}^{(1)}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_2
$cont.$												$\frac{2h\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $\frac{h\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}{}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_1^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_1^\dagger + H_{a_2}S_1H_{a_1}^\dagger S_1^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda_{H1}^{(1)}$	$-\frac{4h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{4h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}S_1S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $+\frac{4h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}S_1S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $+S_1S_{2,j_2}S_1^\dagger S_{2,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$		
$cont.$								$+\frac{2h\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}{}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$		$-\frac{2h\overline{y_{1,i_1,r_1}}\overline{H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}^\dagger S_1^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda_{H1}^{(1)}}}{\epsilon}$ $-h\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}^\dagger S_1^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda_{H1}^{(1)}$	$-\frac{2h\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\epsilon}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1^\dagger S_{2,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	
$cont.$												
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}{}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1,j_2}}\mathcal{I}_{y_{1,a_2,j_1}}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1,j_2}}\mathcal{I}_{y_{1,a_2,j_1}}H_{a_1}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$	$+\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}}{\sqrt{3}\epsilon M_F}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}$ $+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}}{\sqrt{3}M_F}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}$ $+\frac{2h\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$+\frac{2h\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}\lambda_{T3}\overline{I_{y2}{}^*_{a_1,j_1}}\overline{I_{y2}{}^*_{a_3,j_2}}}{\sqrt{3}\epsilon}$ $+\frac{h\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}\lambda_{T3}\overline{I_{y2}{}^*_{a_1,j_1}}\overline{I_{y2}{}^*_{a_3,j_2}}}{\sqrt{3}}$ $-\frac{2h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}}{\epsilon}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\overline{I_{y1}{}_{a_1,j_2}}\overline{I_{y1}{}^*_{a_2,j_1}}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger$ $-\frac{2h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}}{\epsilon}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\overline{I_{y1}{}_{a_1,j_2}}\overline{I_{y1}{}^*_{a_2,j_1}}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}y_{2,i_1,r_1}}\overline{I_{y2}{}_{a_1,j_2}}\overline{I_{y2}{}_{a_2,j_1}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{h\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}y_{2,i_1,r_1}}\overline{I_{y2}{}_{a_1,j_2}}\overline{I_{y2}{}_{a_2,j_1}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{\sqrt{3}}$ $+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$ $+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}}$	$+C\text{NewScalar1NewScalar2}hS_{1,j_1}S_{2,j_2}\text{Bar}(\text{cg1})_{j_1,j_2}$ $+\frac{1}{2}C_{S_1^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(\text{cg1})_{j_1,j_2}$ $+\frac{C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(\text{cg1})_{j_1,j_2}}{\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_2}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_2}\text{Bar}(\text{cg1})_{j_1,j_2}\text{Bar}(\text{cg1})_{j_3,j_4}}{\epsilon}$ $+h\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_2}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_2}\text{Bar}(\text{cg1})_{j_1,j_2}\text{Bar}(\text{cg1})_{j_3,j_4}$ $+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}\text{Bar}(\text{cg1})_{j_2,j_3}S_{1,j_1}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}\text{Bar}(\text{cg1})_{j_2,j_3}S_{1,j_1}{}^\dagger}{\epsilon}$ $+\frac{2h\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}\text{Bar}(\text{cg1})_{j_2,j_3}S_{2,j_1}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{2,i_2,r_2}\text{Bar}(\text{cg1})_{j_2,j_3}S_{2,j_1}{}^\dagger}{\epsilon}$ $+hC\text{NewScalar1NewScalar2cg1}_{j_1,j_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $+\frac{1}{2}C_{S_2^\dagger S_2}h\overline{y_{1,i_1,r_1}}\text{cg1}_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $+\frac{C_{S_1^\dagger S_2}h\overline{y_{1,i_1,r_1}}\text{cg1}_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$ $+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}\text{cg1}_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^\dagger S_{2,j_3}{}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}\text{cg1}_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger S_{2,j_3}{}^\dagger$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}\text{cg1}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger S_{2,j_3}{}^\dagger$ $-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}\text{cg1}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger S_{2,j_3}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $+\frac{h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}}$
cont.													$+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$
													$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}1)_{j_2,j_3}S_{2,j_1}^\dagger \lambda_{S_2}^{(1)}}{\epsilon}$
cont.													$+\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}1)_{j_2,j_3}S_{2,j_1}^\dagger \lambda_{S_2}^{(1)}$ $+\frac{h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}}{\epsilon}$
cont.													$+\frac{1}{2}h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1},j_2}\mathcal{I}_{y_{1,a_2},j_1}^*H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1},j_2}\mathcal{I}_{y_{1,a_2},j_1}^*H_{a_1}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}$	$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^1 \lambda_{12}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^1 \lambda_{12}^{(1)}}{\epsilon}$			
cont.								$-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}$	$+\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^1 \lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^1 \lambda_{12}^{(1)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^1 \lambda_{12}^{(1)}}{\epsilon}$				
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}$	$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^1 \lambda_{12}^{(1)}$ $-\sqrt{3}S_{1,j_2}S_{2,j_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3}S_{1,j_1}S_{2,j_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$				
													$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$+\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$
													$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$
													$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
cont.												$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}{}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$		$-\frac{2h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1,j_2}}\mathcal{I}_{y_{1,a_2,j_1}}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1,j_2}}\mathcal{I}_{y_{1,a_2,j_1}}H_{a_1}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$		
cont.								$+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}{}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$		$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}}{\epsilon}$	$+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$		
cont.										$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$	$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\sqrt{3}S_{1,j_2}S_{2,j_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3}S_{1,j_1}S_{2,j_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$		
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
cont.													$+ \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$+ \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}$
													$- \frac{1}{2} \frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{1}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{2h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$
													$- \frac{10h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$
cont.													$- \frac{10}{3} \frac{h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
													$+ \frac{8}{3} \frac{h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
													$+ \frac{2}{3} \frac{h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}}{\epsilon}$
cont.													$+ \frac{3h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{3}{2} \frac{h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{3h\bar{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{3}{2} \frac{h\bar{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{3h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{3h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{3}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{3}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{3}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	S_1	H	S_1	S_2
												$\begin{aligned} & -\sqrt{3} S_{1,j_2} S_{2,j_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & +\sqrt{3} S_{1,j_1} S_{2,j_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & +\frac{3 h \overline{y}_{1,i_1,r_1}}{\epsilon} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & +\frac{3}{2} \frac{h \overline{y}_{1,i_1,r_1}}{\epsilon} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & +\frac{3 h \overline{y}_{1,i_1,r_1}}{\epsilon} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & -\frac{3}{2} \frac{h \overline{y}_{1,i_1,r_1}}{\epsilon} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & +\frac{3 h \overline{y}_{2,i_1,r_1}}{\epsilon} S_{1,j_2} S_{2,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & +\frac{3}{2} \frac{h \overline{y}_{2,i_1,r_1}}{\epsilon} S_{1,j_2} S_{2,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & -\frac{3 h \overline{y}_{2,i_1,r_1}}{\epsilon} S_{1,j_1} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & -\frac{3}{2} \frac{h \overline{y}_{2,i_1,r_1}}{\epsilon} S_{1,j_1} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & +\frac{2 h \overline{y}_{1,i_1,r_1}}{\sqrt{3} \epsilon} c g^3 j_{2,j_3} S_{1,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)} \\ & +\frac{h \overline{y}_{1,i_1,r_1}}{\epsilon} c g^3 j_{2,j_3} S_{1,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)} \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2 \lambda_{12}^{(2)} \\ & -\frac{\sqrt{3} h \overline{y}_{1,i_1,r_1}}{\epsilon} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2 \lambda_{12}^{(2)} \\ & -\frac{1}{2} \frac{\sqrt{3} h \overline{y}_{1,i_1,r_1}}{\epsilon} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2 \lambda_{12}^{(2)} \\ & -\frac{\sqrt{3} h \overline{y}_{2,i_1,r_1}}{\epsilon} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2 \lambda_{12}^{(2)} \\ & -\frac{1}{2} \frac{\sqrt{3} h \overline{y}_{2,i_1,r_1}}{\epsilon} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2 \lambda_{12}^{(2)} \\ & +\frac{h \overline{y}_{2,i_1,r_1}}{\epsilon} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_1} \text{Bar}(c g^3)_{j_2,j_3} S_{2,j_1}^\dagger \lambda_{S_2}^{(3)} \\ & +\frac{1}{2} \frac{h \overline{y}_{2,i_1,r_1}}{\epsilon} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_1} \text{Bar}(c g^3)_{j_2,j_3} S_{2,j_1}^\dagger \lambda_{S_2}^{(1)} \\ & +\frac{h \overline{y}_{1,i_1,r_1}}{\epsilon} c g^3 j_{1,j_3} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)} \\ & +\frac{1}{2} \frac{h \overline{y}_{1,i_1,r_1}}{\epsilon} c g^3 j_{1,j_3} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)} \end{aligned}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$-\frac{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{3}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$ $+\frac{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}}{\epsilon}$ $+\frac{3}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$
cont.													$-\frac{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}}{\epsilon}$ $-\frac{3}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$
cont.													$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$ $-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$ $-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$ $-\frac{2h\lambda_{T3}L_{a_3,i_1}CP_R\overline{L_{a_4,i_2}}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon M_F}$	$-\frac{4h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3}\epsilon}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3}}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3}}$ $+\frac{H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^{(1)}$ $+\frac{H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^{(2)}$	$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{10h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $-\frac{10}{3}\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{8}{3}\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{2}{3}\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{2}{3}\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(1)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^1\lambda_{12}^{(2)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
cont.										$-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $-\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3M_F}$ $-\frac{2h\overline{\lambda_{T3}L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}^T} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon M_F}$	$-\frac{4h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3}\epsilon}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3}}$	$-\frac{2hy_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $-\frac{10h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3}$ $+\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3}$ $+\frac{8h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3}$ $+\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3}$ $+\frac{2h\overline{\lambda_{T3}L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}^T} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$
cont.									$+\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3}}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3}}$ $+\frac{H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}$ $+\frac{H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}}{\epsilon}$	$-\frac{10h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $-\frac{10h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3}$ $+\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3}$ $+\frac{8h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3}$ $+\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3}$ $+\frac{2h\overline{\lambda_{T3}L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}^T} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$
cont.									$-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}$ $+\frac{H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}$	$-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	S_1	S_2
$cont.$								$-\frac{h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^2 \lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{2h\bar{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}^1 \lambda_{12}^{(1)}}{\epsilon}$	
								$-\frac{2\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$	$-\frac{2h\bar{y}_{1,i_1,r_1}V_{e,j_2,j_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,j_3}H_{a_3}^\dagger S_{1,j_1}^\dagger}{\sqrt{3\epsilon}}$	$-\frac{4h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$-\frac{2h\bar{y}_{1,i_1,r_1}V_{e,j_2,j_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,j_3}H_{a_3}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$-\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
								$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon M_F}$	$-\frac{2i\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1}Y_{e,i_2,j_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,j_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^1}{\epsilon}$	$-\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
$cont.$								$-2i\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1}Y_{e,i_2,j_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,j_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^1$	$-\frac{4h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon}$	$-\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
								$+\frac{H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^2 \lambda_{H1}^{(2)}}{\sqrt{3\epsilon}}$	$-\frac{4h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon}$	$+\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
$cont.$								$-\frac{h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^2 \lambda_{H1}^{(2)}}{\sqrt{3}}$	$+\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$+\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
									$+\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$+\frac{4}{3}h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
$cont.$									$-\frac{2h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1 \lambda_{12}^{(1)}}{\sqrt{3\epsilon}}$	$-\frac{2h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$
									$-\frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1 \lambda_{12}^{(1)}}{\sqrt{3\epsilon}}$
$cont.$									$-\frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1 \lambda_{12}^{(1)}}{\sqrt{3\epsilon}}$	$-\frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$
									$-\frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1 \lambda_{12}^{(1)}}{\sqrt{3\epsilon}}$
$cont.$									$-\frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1 \lambda_{12}^{(1)}}{\sqrt{3\epsilon}}$
									$-\frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\$	

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}e}$
cont.													$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}e}$ $+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.													$-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}e}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}$
cont.													$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}e}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}e}$
cont.													$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$

Field combinations 4 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	$S_1 \ell$	S_2	H
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$			$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_1^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_1^\dagger \right)^2$	$-\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$	
										$-2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1 y_{1,i_1,r_1} S_1^\dagger$	$-\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$	
										$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$	$+\frac{2i\sqrt{\frac{2}{3}} h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}}{\epsilon}$	
cont.											$+2i\sqrt{\frac{2}{3}} h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}$	
cont.											$-\frac{2h\overline{y_{2,i_1,r_1} H_{a_2} S_{2,j_2} \overline{y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1} H_{a_2} S_{2,j_2} \overline{y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}}$	
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0			$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_1^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_1^\dagger \right)^2$	$-\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$	
										$-2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1 y_{1,i_1,r_1} S_1^\dagger$	$-\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$	
										$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$	$+\frac{2i\sqrt{\frac{2}{3}} h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}}{\epsilon}$	
cont.											$+2i\sqrt{\frac{2}{3}} h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}$	
cont.											$-\frac{2h\overline{y_{2,i_1,r_1} H_{a_2} S_{2,j_2} \overline{y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1} H_{a_2} S_{2,j_2} \overline{y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}}$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$			$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_1^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_1^\dagger \right)^2$	$-\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$	
										$-2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1 y_{1,i_1,r_1} S_1^\dagger$	$-\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$	
										$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$	$+\frac{2i\sqrt{\frac{2}{3}} h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}}{\epsilon}$	
cont.											$+2i\sqrt{\frac{2}{3}} h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}$	
cont.											$-\frac{2h\overline{y_{2,i_1,r_1} H_{a_2} S_{2,j_2} \overline{y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1} H_{a_2} S_{2,j_2} \overline{y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}}$	
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1			$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_1^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_1^\dagger \right)^2$	$-\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$	
										$-2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1 y_{1,i_1,r_1} S_1^\dagger$	$-\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$	
										$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$	$+\frac{2i\sqrt{\frac{2}{3}} h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}}{\epsilon}$	
cont.											$+2i\sqrt{\frac{2}{3}} h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}$	
cont.											$-\frac{2h\overline{y_{2,i_1,r_1} H_{a_2} S_{2,j_2} \overline{y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1} H_{a_2} S_{2,j_2} \overline{y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}}$	
T3-A	2	1	1	3	2	2	$\frac{3}{2}$			$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_1^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_1^\dagger \right)^2$	$-\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$	
										$-2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1 y_{1,i_1,r_1} S_1^\dagger$	$-\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$	
										$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$	$+\frac{2i\sqrt{\frac{2}{3}} h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}}{\epsilon}$	
cont.											$+2i\sqrt{\frac{2}{3}} h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} \overline{y_{2,i_2,r_1} Y_{e,i_1,i_3}} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}$	
cont.											$-\frac{2h\overline{y_{2,i_1,r_1} H_{a_2} S_{2,j_2} \overline{y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1} H_{a_2} S_{2,j_2} \overline{y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F$	$S_1 \ell$	$S_2 H$
cont.										$-\frac{h\overline{y}_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^2 \lambda_{H2}^{(2)}}{\sqrt{3}}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-C_{S_1^\dagger S_1} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}{}^\dagger \lambda_{S_1}^{(1)}}$		$-\frac{2h\overline{y}_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y2,a_1,j_1}\mathcal{I}_{y2,a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\epsilon}$ $-2h\overline{y}_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y2,a_1,j_1}\mathcal{I}_{y2,a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger \lambda_{H2}^{(1)}$ $-\frac{h\overline{y}_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger \lambda_{H2}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y}_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger \lambda_{H2}^{(1)}$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger \lambda_{H2}^{(2)}$ $-\frac{h\overline{y}_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger \lambda_{H2}^{(2)}}{\epsilon}$ $-\frac{1}{2}h\overline{y}_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger \lambda_{H2}^{(2)}$
cont.										
cont.										
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-C_{S_1^\dagger S_1} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}{}^\dagger \lambda_{S_1}^{(1)}}$	$\frac{3h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_1} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2} g_2^2}{8eM_F}$ $+\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_1} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2} g_2^2}{8M_F}$ $+3h\overline{y_{1,i_1,r_1} y_{1,i_2,r_1} L_{a_2,i_2}}{}^T CPL_{a_1,i_1} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger g_2^2$ $+\frac{8eM_F}{8eM_F}$ $+h\overline{y_{1,i_1,r_1} y_{1,i_2,r_1} L_{a_2,i_2}}{}^T CPL_{a_1,i_1} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger g_2^2$ $+\frac{8M_F}{8M_F}$ $+3h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_1} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}$ $+\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_1} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}}{8M_F}$ $+\frac{\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_1} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}}{2M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{1,i_3,r_1} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}}{2eM_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{1,i_3,r_1} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}}{4M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{1,i_3,r_2} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}}{4eM_F}$ $-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{1,i_3,r_2} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}}{8M_F}$ $-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{1,i_3,r_1} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}}{4eM_F}$ $-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{1,i_3,r_2} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}}{8M_F}$ $-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{1,i_3,r_2} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}}{eM_F}$ $-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{1,i_3,r_2} \mathcal{I}_{y1,a_1,j_1} \mathcal{I}_{y1,a_2,j_2}}{2M_F}$ $-\frac{h\overline{y_{1,i_1,r_$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	S_1	ℓ	S_2	H
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-C_{S_1^\dagger S_1} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}$				$-\frac{2h\overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y_2,a_1,j_1} T_{y_{2,a_2,j_2}}^* H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$	
								$-\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$				$-\frac{2h\overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y_2,a_1,j_1} T_{y_{2,a_2,j_2}}^* H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$	
								$-\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1,j_3,j_2,j_4}} \lambda_{S_1}^{(1)}}$				$+\frac{H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} \lambda_{H2}^{(1)}}{\epsilon}$	
cont.												$-\frac{h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} \lambda_{H2}^{(1)}}{\epsilon}$	
cont.												$-\frac{1}{2} h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} \lambda_{H2}^{(1)}}{2}$	
												$+\frac{H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} \lambda_{H2}^{(2)}}{2}$	
												$-\frac{h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} \lambda_{H2}^{(2)}}{\epsilon}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-C_{S_1^\dagger S_1} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}$				$-\frac{2h\overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y_2,a_1,j_1} T_{y_{2,a_2,j_2}}^* H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$	
								$-\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$				$-\frac{2h\overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y_2,a_1,j_1} T_{y_{2,a_2,j_2}}^* H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$	
								$-\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1,j_3,j_2,j_4}} \lambda_{S_1}^{(1)}}$				$+\frac{H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} \lambda_{H2}^{(1)}}{\epsilon}$	
cont.												$-\frac{h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} \lambda_{H2}^{(1)}}{\epsilon}$	
cont.												$-\frac{1}{2} h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} \lambda_{H2}^{(1)}}{2}$	
												$+\frac{H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} \lambda_{H2}^{(2)}}{2}$	
												$-\frac{h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2} \lambda_{H2}^{(2)}}{\epsilon}$	
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-C_{S_1^\dagger S_1} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}$				$-\frac{2h\overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y_2,a_1,j_1} T_{y_{2,a_2,j_2}}^* H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	S_1	ℓ	S_2	H
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\sqrt{3}C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
								$-\frac{5}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$
								$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_2,j_4}} \lambda_{S_1}^{(1)}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$
cont.								$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$
cont.								$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$
cont.								$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H,a_1,a_2,j_1,j_2}}{\sqrt{3}\epsilon}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} (S_{1,j_1}^\dagger)^2}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$
								$-\frac{2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}}$					$-2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$
								$-\frac{4}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$					$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_4,j_2}} \lambda_{S_1}^{(1)}}{\sqrt{3}}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}}{\epsilon}$
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_4,j_2}} \lambda_{S_1}^{(2)}}{\sqrt{3}}$					$-\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+\frac{2}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} (S_{1,j_1}^\dagger)^2}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$
								$-\frac{2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}}$					$-2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$
								$-\frac{4}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$					$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_4,j_2}} \lambda_{S_1}^{(1)}}{\sqrt{3}}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}}{\epsilon}$
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_4,j_2}} \lambda_{S_1}^{(2)}}{\sqrt{3}}$					$-\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} (S_{1,j_1}^\dagger)^2}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$
								$-\frac{2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}}$					$-2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$
								$-\frac{4}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$					$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_4,j_2}} \lambda_{S_1}^{(1)}}{\sqrt{3}}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}}{\epsilon}$
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_4,j_2}} \lambda_{S_1}^{(2)}}{\sqrt{3}}$					$-\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+\frac{2}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} (S_{1,j_1}^\dagger)^2}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$
								$-\frac{2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}}$					$-2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$
								$-\frac{4}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$					$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_4,j_2}} \lambda_{S_1}^{(1)}}{\sqrt{3}}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}}{\epsilon}$
cont.								$-\frac{2\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_2,j_4,j_2}} \lambda_{S_1}^{(2)}}{\sqrt{3}}$					$-\hbar \overline{y_{2,i_1,r_1} H_{a_2} S_{2,y_2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} (S_{1,j_1}^\dagger)^2}$					$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$
								$-\frac{2C_{S_1^\dagger S_1} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}}$					$-2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,y_2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$
								$-\frac{4}{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$					$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}} \lambda_{H2}^{(1)}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	S_1	ℓ	S_2	H
$cont.$													$-\frac{\overline{hy_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger T_{H^2,a_1,a_2,j_2,j_1}}{\sqrt{3}}{}^2\lambda_{H^2}^{(2)}$

Field combinations 5 of 10

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 F$	$S_2 \ell$	$F B_{\mu\nu}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger \\ & -\frac{5}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S_2,j_1,j_3,j_2,j_4}^{(1)} \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & + \frac{3h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2} g_2^2}{8 M_F} \\ & + \frac{h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2} g_2^2}{8 M_F} \\ & + \frac{3h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T \text{cg}1_{a_1,j_1,a_2,j_2} y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger g_2^2}{8 M_F} \\ & + \frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T \text{cg}1_{a_1,j_1,a_2,j_2} y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger g_2^2}{8 M_F} \\ & + \frac{3h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_3,i_3} g_1^2 S_{2,j_1} S_{2,j_2} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2}}{8 M_F} \\ & + \frac{h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1} S_{2,j_2} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2}}{8 M_F} \\ & + \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2}}{2 M_F} \\ & - \frac{\sqrt{3} h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} y_{2,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{1,i_3,r_1} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2}}{4 M_F} \\ & - \frac{3\sqrt{3} h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} y_{2,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{1,i_3,r_1} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2}}{8 M_F} \\ & - \frac{\sqrt{3} h \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2}}{4 M_F} \\ & - \frac{3\sqrt{3} h \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2}}{8 M_F} \\ & - \frac{\sqrt{3} h \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} y_{2,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2}}{2 M_F} \\ & - \frac{\sqrt{3} h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} y_{2,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(\text{cg}1)_{a_1,j_1,a_2,j_2}}{4 M_F} \\ & + \frac{3h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T \text{cg}1_{a_1,j_1,a_2,j_2} g_1^2 y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{8 M_F} \\ & + \frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T \text{cg}1_{a_1,j_1,a_2,j_2} g_1^2 y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{8 M_F} \\ & + \frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T \text{cg}1_{a_1,j_1,a_2,j_2} y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{2 M_F} \end{aligned}$	
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T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger \\ & -\frac{5}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S_2,j_1,j_3,j_2,j_4}^{(1)} \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & -\frac{3}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned}$	
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger \\ & -\frac{5}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S_2,j_1,j_3,j_2,j_4}^{(1)} \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & -3 \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned}$	
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger \\ & -\frac{5}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S_2,j_1,j_3,j_2,j_4}^{(1)} \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & -\frac{27}{4} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\begin{aligned} & -2 \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_2}^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger \right)^2 \\ & -2 \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & -\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	F	S_2	ℓ	F	$B_{\mu\nu}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger \right)^2$					
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_2} y_{2,i_1,r_1} S_2^\dagger$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \varsigma_2^2 y_{2,i_1,r_1} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger \right)^2$					$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_2} y_{2,i_1,r_1} S_2^\dagger$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \varsigma_2^2 y_{2,i_1,r_1} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger \right)^2$					$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_2} y_{2,i_1,r_1} S_2^\dagger$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \varsigma_2^2 y_{2,i_1,r_1} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger \right)^2$					$-\frac{9}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_2} y_{2,i_1,r_1} S_2^\dagger$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \varsigma_2^2 y_{2,i_1,r_1} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger \right)^2$					$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$
								$-\frac{2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1}} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}}$					
								$-\frac{\frac{4}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger + 2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4}} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(1)}}{\sqrt{3}}$					
cont.								$2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4}} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(2)}$					
								$-\frac{\sqrt{3}}{\sqrt{3}}$					
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{2}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger \right)^2$					
								$-\frac{2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1}} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}}$					
								$-\frac{\frac{4}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger + 2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4}} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(1)}}{\sqrt{3}}$					
cont.								$2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4}} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(2)}$					
								$-\frac{\sqrt{3}}{\sqrt{3}}$					
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger \right)^2$					$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$
								$-\frac{2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1}} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}}$					
								$-\frac{\frac{4}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger + 2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4}} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(1)}}{\sqrt{3}}$					
cont.								$2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4}} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(2)}$					
								$-\frac{\sqrt{3}}{\sqrt{3}}$					

[illegible]

Field combinations 6 of 10

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H\ell e$	$S_1 S_2 H$
									$+\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{3}{8}\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{8}$	$-\frac{1}{4}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{4}$
										$-\frac{\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.										$-\frac{1}{4}\frac{\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{4}$
									$+\frac{\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\lambda\overline{T_3}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1		$+\frac{3}{8}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
									$+\frac{\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{4}$
									$+\frac{3}{8}\frac{\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{8}$	$-\frac{\hbar y_{2,i_1,r_1}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4}\frac{\hbar y_{2,i_1,r_1}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{4}$
									$+\frac{3}{8}\frac{\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{8}$	$+\frac{H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{4}$
									$+\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{3}{8}\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{8}$	$-\frac{1}{4}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{4}$
										$-\frac{\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.										$-\frac{1}{4}\frac{\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{4}$
									$+\frac{\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\lambda\overline{T_3}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$		$+\frac{3}{8}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
									$+\frac{\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{4}$
									$+\frac{3}{8}\frac{\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{8}$	$-\frac{\hbar y_{2,i_1,r_1}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4}\frac{\hbar y_{2,i_1,r_1}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{4}$
									$+\frac{3}{8}\frac{\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{8}$	$+\frac{H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{4}$
									$+\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{3}{8}\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{8}$	$-\frac{1}{4}\frac{\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{4}$
										$-\frac{\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.										$-\frac{1}{4}\frac{\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{4}$
									$+\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\lambda\overline{T_3}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{3}{8}\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{8}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
									$+\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{8}$	$-\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4}\sqrt{3}\hbar y_{2,i_1,r_1}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8}\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{8}$	$+\frac{H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{4}$
									$+\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{8}$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
										$-\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW\mu\nu$	$H\ell e$	S_1S_2H
<i>cont.</i>										$-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}\\&+\frac{\sqrt{3h\bar{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}\end{aligned}$	$\begin{aligned}&-\frac{2h\bar{y}_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger}{\epsilon}\\&-\frac{2h\bar{y}_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger}{\epsilon+\lambda T_3S_{2,j_2}T_3^{*a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}\end{aligned}$
<i>cont.</i>									$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger\end{aligned}$	$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>									$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger\end{aligned}$	$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&\quad +H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{2h\bar{y}_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}cg^4_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{2h\bar{y}_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}cg^4_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\bar{y}_{1,i_1,r_1}H_{a_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(cg3)_{j_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(1)}}\\&\quad -\frac{1}{2}h\bar{y}_{2,i_1,r_1}\overline{H_{a_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(cg3)_{j_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(1)}}\\&-\frac{3h\bar{y}_{2,i_1,r_1}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(1)}}{\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{3}{2}h\bar{y}_{2,i_1,r_1}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(1)}\\&-\frac{h\bar{y}_{1,i_1,r_1}cg3_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H_2}^{(1)}}{\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{1}{2}h\bar{y}_{1,i_1,r_1}cg3_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H_2}^{(1)}\\&-\frac{3h\bar{y}_{1,i_1,r_1}cg4_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H_2}^{(1)}}{\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{3}{2}h\bar{y}_{1,i_1,r_1}cg4_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H_2}^{(1)}\\&+\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(2)}}{\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&+\frac{1}{2}\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(cg4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger\lambda_{H_2}^{(2)}\\&+\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}cg4_{a_1,j_1,a_2,j_2}H_{a_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H_2}^{(2)}}{\epsilon}\end{aligned}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}\\&+\frac{\sqrt{3h\bar{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}\end{aligned}$	$\begin{aligned}&\quad +\overline{\lambda T_3}S_{2,j_2}T_3^{*a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}\\&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}\\&-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&\quad +H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\\&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>									$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger\end{aligned}$	$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>									$\begin{aligned}&+\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}\\&+\frac{3}{8}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger\end{aligned}$	$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger\\&\quad +H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\end{aligned}$
<i>cont.</i>										$\begin{aligned}&-\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger\\&-\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}\end{aligned}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H\ell e$	S_1S_2H
cont.									$+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $+\frac{\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
									$+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H\ell e$	$S_1 S_2 H$
cont.									$+\frac{3}{8}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $-\frac{1}{2}\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{1}{2}\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$+\overline{\lambda_{T3}}S_{2,j_2}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $+\frac{H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
									$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H \ell e$	$S_1 S_2 H$
cont.									$+\frac{3}{8}\sqrt{3h\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{h\overline{y_{2,i_1,r_1}\overline{\lambda_{T3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}}{\sqrt{3\epsilon}}$ $-\frac{h\overline{y_{2,i_1,r_1}\overline{\lambda_{T3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}}{2\sqrt{3}}$
									$+\frac{3}{8}\sqrt{3h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger$ $-\frac{h\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}}{\sqrt{3\epsilon}}$ $-\frac{h\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}}{2\sqrt{3}}$
cont.									$+\frac{3}{8}\sqrt{3h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{h\overline{y_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}}{\sqrt{3\epsilon}}$ $-\frac{h\overline{y_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}}{2\sqrt{3}}$
cont.									$+\frac{\sqrt{3h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3h\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3h\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$+\overline{\lambda_{T3}S_{2,j_2}\mathcal{I}T_{3a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}$ $-\frac{h\overline{y_{1,i_1,r_1}\overline{\lambda_{T3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}}{\sqrt{3\epsilon}}$ $-\frac{h\overline{y_{1,i_1,r_1}\overline{\lambda_{T3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}}{2\sqrt{3}}$ $-\frac{h\overline{y_{2,i_1,r_1}\overline{\lambda_{T3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}}{\sqrt{3\epsilon}}$ $-\frac{h\overline{y_{2,i_1,r_1}\overline{\lambda_{T3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}}{2\sqrt{3}}$ $+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger$ $-\frac{h\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}}{\sqrt{3\epsilon}}$ $-\frac{h\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}}{2\sqrt{3}}$ $-\frac{h\overline{y_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}}{\sqrt{3\epsilon}}$ $-\frac{h\overline{y_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}}{2\sqrt{3}}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3h\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3h\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$+\overline{\lambda_{T3}S_{2,j_2}\mathcal{I}T_{3a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}$ $-\frac{h\overline{y_{1,i_1,r_1}\overline{\lambda_{T3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}}{\sqrt{3\epsilon}}$

Field combinations 7 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{4\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_2,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$	$+\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{M_F}$	$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_2} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger$
								$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{12,j_2,j_1}^1 \lambda_{12}^{(1)}$	$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4\epsilon M_F}$	
								$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{12,j_2,j_1}^1 \lambda_{12}^{(1)}$	$-\frac{3\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{8M_F}$	
cont.								$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{\epsilon M_F}$	$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{2M_F}$	
cont.								$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4\epsilon M_F}$	$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{8M_F}$	
cont.								$-\frac{3\sqrt{3}\hbar \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{8M_F}$	$-\frac{3\sqrt{3}\hbar \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{8M_F}$	
cont.								$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{2\sqrt{3}M_F}$	
cont.								$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	$-\frac{3\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{8M_F}$	
cont.								$+\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{M_F}$	$-\frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} y_{2,i_2,i_1,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
cont.								$-\frac{3\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} y_{2,i_2,i_1,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{8M_F}$	$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{\epsilon M_F}$	
cont.								$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,i_2,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{2M_F}$	$-\frac{\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{\sqrt{3}\epsilon M_F}$	
cont.								$-\frac{\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{2\sqrt{3}M_F}$	$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_2,r_2} y_{2,i_2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
cont.								$-\frac{3\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} y_{2,i_2,i_1,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{8M_F}$	$-\frac{3\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_2,r_2} y_{2,i_2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{8M_F}$	
cont.								$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_2,r_2} y_{2,i_2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	$-\frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,i_2,r_2} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
cont.								$-\frac{3\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,i_2,r_2} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{8M_F}$	$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.								$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger \lambda_{12}^{(1)}}{M_F}$	$-\frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.								$-\frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$		
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{4\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_2,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$	$3\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger g_Y^2$	$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_2} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger$
								$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{12,j_2,j_1}^1 \lambda_{12}^{(1)}$	$+\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger g_Y^2}{4M_F}$	
								$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{12,j_2,j_1}^1 \lambda_{12}^{(1)}$	$+\frac{3\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger g_Y^2}{4\epsilon M_F}$	
cont.								$+\frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger g_Y^2}{4M_F}$	$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{M_F}$	
cont.								$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4\epsilon M_F}$	$-\frac{\hbar \overline{y_{1,i_1,r_1}} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger S_1^\dagger \mathcal{I} H_{1,a_1,a_2}^{-1} \lambda_{H1}^{(1)}}{M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$-\frac{3 h \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
									$-\frac{h \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
cont.									$-\frac{h \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{4 \epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
									$-\frac{h \overline{y}_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{4 \epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
cont.									$-\frac{3 h \overline{y}_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
									$-\frac{\sqrt{3} h \overline{y}_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{4 \epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
cont.									$-\frac{3 \sqrt{3} h \overline{y}_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
									$-\frac{h \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} L_{a_2,i_2}}{\sqrt{3} \epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
cont.									$-\frac{h \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}}{2 \sqrt{3} M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
									$-\frac{h \overline{y}_{1,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{4 \epsilon M_F} T S_1 y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
cont.									$-\frac{3 h \overline{y}_{1,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{8 M_F} T S_1 y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
									$+\frac{\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{M_F} T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
cont.									$-\frac{\sqrt{3} h \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{4 \epsilon M_F} T S_1 y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
									$-\frac{3 \sqrt{3} h \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{8 M_F} T S_1 y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
cont.									$-\frac{h \overline{y}_{1,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{\epsilon M_F} T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
									$-\frac{h \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{2 M_F} T S_1 y_{1,i_3,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
cont.									$-\frac{h \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{\sqrt{3} \epsilon M_F} T S_1 y_{1,i_3,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
									$-\frac{h \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{2 \sqrt{3} M_F} T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
cont.									$-\frac{h \overline{y}_{1,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{4 \epsilon M_F} T S_1 y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
									$-\frac{3 h \overline{y}_{1,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{8 M_F} T S_1 y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
cont.									$-\frac{\sqrt{3} h \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{4 \epsilon M_F} T S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
									$-\frac{3 \sqrt{3} h \overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{8 M_F} T S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
cont.									$-\frac{h \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}{\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} I_{T3,a_1,a_2,j_1} S_1^\dagger \lambda_{12}^{(1)}$	
									$-\frac{h \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}{M_F} T C P_L L_{a_1,i_1} S_{2,j_1} I_{T3,a_1,a_2,j_1} S_1^\dagger \lambda_{12}^{(1)}$	
cont.									$-\frac{h \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{\epsilon M_F} T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}$	
									$-\frac{h \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{M_F} T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} I_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{4 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$	$+\frac{\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}{M_F} T C P_L L_{a_1,i_1} S_{2,j_1} I_{T3,a_1,a_2,j_1} S_1^\dagger$	$-2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger$
								$-2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger I_{12,j_2,j_1}^\dagger \lambda_{12}^{(1)}$	$-\frac{h \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{4 \epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} I_{T3,a_1,a_2,j_1} S_1^\dagger$	$-2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger S_1^\dagger I_{H1,a_1,a_2}^\dagger \lambda_{H1}^{(1)}$
								$-\frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger I_{12,j_2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{3 h \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
cont.									$-\frac{h \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
									$-\frac{h \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{2 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
cont.									$-\frac{h \overline{y}_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{4 \epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
									$-\frac{3 h \overline{y}_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
cont.									$-\frac{\sqrt{3} h \overline{y}_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{4 \epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
									$-\frac{3 \sqrt{3} h \overline{y}_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
cont.									$-\frac{\sqrt{3} h \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{4 \epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
									$-\frac{3 \sqrt{3} h \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
cont.									$-\frac{h \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{\sqrt{3} \epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	
									$-\frac{h \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{\sqrt{3} \epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_1} S_1^\dagger$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}{2\sqrt{3}M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{4\epsilon M_F}$			
													$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{8M_F} + \frac{\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{M_F}$			
cont.													$-\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_3,r_2}y_{2,i_4,r_2}y_{2,i_2,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{4\epsilon M_F}$			
													$-\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_3,r_2}y_{2,i_4,r_2}y_{2,i_2,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{8M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{\epsilon M_F}$			
cont.													$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{2M_F}$			
													$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{\sqrt{3}\epsilon M_F}$			
													$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{2\sqrt{3}M_F}$			
cont.													$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{4\epsilon M_F}$			
													$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{8M_F}$			
													$-\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{8M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger\lambda_{12}^{(1)}}}{\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger\lambda_{12}^{(1)}}}{M_F}$			
cont.													$-\frac{h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$-\frac{h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger\lambda_{12}^{(1)}}{M_F}$			
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1						$-\frac{4h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$			
													$-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}S_1S_{2,j_2}y_{1,i_1,r_1}S_1}^\dagger S_{2,j_1}^\dagger\overline{\mathcal{I}T_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}$			
													$-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}S_1S_{2,j_2}y_{2,i_1,r_1}S_1}^\dagger S_{2,j_1}^\dagger\overline{\mathcal{I}T_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}$			
cont.													$-\frac{9h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}g_Y^2S_{2,j_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{4\epsilon M_F}$			
													$-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}g_Y^2S_{2,j_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{4M_F}$			$-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{c,i_2,i_3}H_{a_1}S_1y_{1,i_2,r_1}Y_{c,i_1,i_3}H_{a_1}}^\dagger S_1^\dagger$
													$+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{M_F}$			$-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}}^\dagger S_1^\dagger\overline{\mathcal{I}H_{1,a_1,a_2}{}^1\lambda_{H1}^{(1)}}$
cont.													$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{4\epsilon M_F}$			
													$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{8M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{\epsilon M_F}$			
cont.													$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{2M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{4\epsilon M_F}$			
													$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{8M_F}$			
cont.													$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{4\epsilon M_F}$			
													$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{8M_F}$			
													$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{4\epsilon M_F}$			
cont.													$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{8M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{\sqrt{3}\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}{}^TC P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_1}^\dagger}}{2\sqrt{3}M_F}$			
cont.													$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}}{4\epsilon M_F}$			
													$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_4,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}}{8M_F}$			
													$+\frac{\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}{M_F}$			
cont.													$-\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}{}^TS_1y_{1,i_3,r_2}y_{2,i_4,r_1}y_{2,i_2,r_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1}S_2}}^\dagger}}{4\epsilon M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$-\frac{3\sqrt{3}\overline{h\overline{y_2}}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{\epsilon M_F}$			
cont.													$-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{2M_F}$			
													$-\frac{h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon M_F}$			
cont.													$-\frac{h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{2\sqrt{3}M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3h\overline{y_{1,i_3,r_1}}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{\sqrt{3}h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3\sqrt{3}h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
cont.													$-\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}}{M_F}$			
													$-\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
cont.													$-\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
													$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{L}_{a_1,i_1}}{8M_F}CPr\overline{L}_{a_2,i_2}^*S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{I}T3_{a_1,a_2,j_1}^*S_{2,j_1}^\dagger$ $-\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{4M_F}CPr\overline{L}_{a_2,i_2}^*S_1y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\overline{I}T3_{a_1,a_2,j_1}^*S_{2,j_1}^\dagger$ $-\frac{3\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}CPr\overline{L}_{a_2,i_2}^*S_1y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\overline{I}T3_{a_1,a_2,j_1}^*S_{2,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{L}_{a_2,i_2}}{\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_1}\overline{I}T3_{a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}$			
cont.													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{L}_{a_2,i_2}}{M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_1}\overline{I}T3_{a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}$ $-\frac{h\overline{L}_{a_1,i_1}}{M_F}CPr\overline{L}_{a_2,i_2}^*S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{I}T3_{a_1,a_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}$ $-\frac{\epsilon M_F}{h\overline{L}_{a_1,i_1}}CPr\overline{L}_{a_2,i_2}^*S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{I}T3_{a_1,a_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.																
cont.																
													$+2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y}_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\overline{I}T_{12,j_2,j_3,j_1,j_4}^{(1)}\lambda_{12}^{(1)}$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y}_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\overline{I}T_{12,j_2,j_3,j_1,j_4}^{(1)}\lambda_{12}^{(1)}$			
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$						$+3h\overline{y}_{1,i_2,r_1}\overline{y}_{2,i_1,r_1}\overline{L}_{a_2,i_2}^*CPL\overline{L}_{a_1,i_1}g_2^2S_{2,j_2}\overline{I}y_{1,a_1,j_1}\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $+\frac{h\overline{y}_{1,i_2,r_1}\overline{y}_{2,i_1,r_1}\overline{L}_{a_2,i_2}}{4M_F}^TCPL\overline{L}_{a_1,i_1}g_2^2S_{2,j_2}\overline{I}y_{1,a_1,j_1}\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $+\frac{\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{L}_{a_2,i_2}}{M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}\overline{I}y_{1,a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{8M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{2\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{8M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{I}y_{a_1,j_1}^*\overline{I}y_{2,a_2,j_2}^*S_{1,j_1}^\dagger$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}\overline{L}_{a_2,i_2}}{4\epsilon M_F}^TCPL\overline{L}_{a_1,i_1}S_{2,j_$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$+ \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} g_{2, i_1, i_1}^2 S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$ $+ \frac{h \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$ $+ \frac{h \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$ $+ \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$ $+ \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$- \frac{h \overline{y_{1, i_2, r_1}} y_{2, i_1, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I}_{y_{1, a_1, j_1}}^* \mathcal{I}_{y_{2, a_2, j_2}} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$ $- \frac{h \overline{y_{1, i_2, r_1}} y_{2, i_1, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I}_{y_{1, a_1, j_1}}^* \mathcal{I}_{y_{2, a_2, j_2}} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$ $- \frac{h \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I}_{y_{1, a_1, j_1}}^* \mathcal{I}_{y_{2, a_2, j_2}} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$ $- \frac{h \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I}_{y_{1, a_1, j_1}}^* \mathcal{I}_{y_{2, a_2, j_2}} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$ $- \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \mathcal{I}_{y_{1, a_1, j_2}} \mathcal{I}_{y_{2, a_2, j_1}} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$ $- \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \mathcal{I}_{y_{1, a_1, j_2}} \mathcal{I}_{y_{2, a_2, j_1}} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$ $- \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \mathcal{I}_{y_{1, a_1, j_2}} \mathcal{I}_{y_{2, a_2, j_1}} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$ $- \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \mathcal{I}_{y_{1, a_1, j_2}} \mathcal{I}_{y_{2, a_2, j_1}} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
cont.									$- \frac{h \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I}_{y_{1, a_1, j_1}}^* \mathcal{I}_{y_{2, a_2, j_2}} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$ $- \frac{h \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I}_{y_{1, a_1, j_1}}^* \mathcal{I}_{y_{2, a_2, j_2}} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$ $- \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \mathcal{I}_{y_{1, a_1, j_2}} \mathcal{I}_{y_{2, a_2, j_1}} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$ $- \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \mathcal{I}_{y_{1, a_1, j_2}} \mathcal{I}_{y_{2, a_2, j_1}} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$ $- \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \mathcal{I}_{y_{1, a_1, j_2}} \mathcal{I}_{y_{2, a_2, j_1}} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$ $- \frac{h \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \mathcal{I}_{y_{1, a_1, j_2}} \mathcal{I}_{y_{2, a_2, j_1}} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
cont.									$+ \frac{C_{S_1^2 S_2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right)}{\sqrt{2} S_{1, r_1} S_{1, j_1} S_{2, j_2} y_{1, i_1, r_1} \text{Bar}(\text{eg} 1)_{j_1, j_2}} + \frac{3 h \overline{y_{1, i_2, r_1}} y_{2, i_1, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I}_{y_{1, a_1, j_1}}^* \mathcal{I}_{y_{2, a_2, j_2}} S_{1, j_1}^\dagger g_2^2}{4 \epsilon M_F}$ $- h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1, i_1, r_1}} y_{2, i_2, r_2} S_{1, j_1} S_{1, j_2} S_{2, j_2} S_{2, j_4} y_{1, i_1, r_2} y_{1, i_2, r_1} \text{Bar}(\text{eg} 1)_{j_1, j_2} \text{Bar}(\text{eg} 1)_{j_2, j_4}} + \frac{h \overline{y_{1, i_2, r_1}} y_{2, i_1, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I}_{y_{1, a_1, j_1}}^* \mathcal{I}_{y_{2, a_2, j_2}} S_{1, j_1}^\dagger g_2^2}{4 M_F}$ $+ 2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1, i_1, r_1}} y_{2, i_2, r_2} S_{1, j_1} S_{1, j_2} S_{2, j_3} y_{1, i_1, r_2} y_{1, i_2, r_1} \text{Bar}(\text{eg} 1)_{j_2, j_3} S_{1, j_1}^\dagger} + \frac{3 h \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger g_2^2}{4 M_F}$ $+ 2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1, i_1, r_1}} y_{2, i_2, r_2} S_{1, j_2} S_{2, j_1} S_{2, j_3} y_{1, i_1, r_2} y_{2, i_2, r_1} \text{Bar}(\text{eg} 1)_{j_2, j_3} S_{2, j_1}^\dagger} + \frac{h \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger g_2^2}{4 M_F}$ $+ C_{S_1^2 S_2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1, i_1, r_1}} \text{eg} 1_{j_1, j_2$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}\overline{y}_{2,i_3,r_3}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}I_{y_{1,a_1,j_1}}^* I_{y_{2,a_2,j_2}}^* S_{1,j_1}^\dagger}{8M_F}$	
									$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}I_{y_{1,a_1,j_1}}^* I_{y_{2,a_2,j_2}}^* S_{1,j_1}^\dagger}{2\epsilon M_F}$	
									$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}I_{y_{1,a_1,j_1}}^* I_{y_{2,a_2,j_2}}^* S_{1,j_1}^\dagger}{4M_F}$	
									$+ \frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_2^2 S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$+ \frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_2^2 S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4M_F}$	
									$+ \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_2^2 S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$+ \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_2^2 S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{4M_F}$	
									$+ \frac{\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_3,r_1}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{M_F}$	
cont.									$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{2\epsilon M_F}$	
									$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{8M_F}$	
									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{3h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{8M_F}$	
									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_3,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_3,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{3h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_3,r_2}I_{y_{1,a_1,j_2}}I_{y_{2,a_2,j_1}}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$+ \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4M_F}$	
									$+ \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.										$+\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$			$\frac{3h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{8M_F}$			
										$-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$			$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{24M_F}$			
cont.										$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$			$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.										$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$			$\frac{3h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{3h\overline{y_{1,i_3,r_1}}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{3h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{8M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_{1,i_1,j_1}}{}^*I_{y_{2,a_2,j_2}}{}^*S_{1,j_1}^\dagger}{4M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$	
cont.									$-\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}T_{y_1,a_1,j_2}T_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3\epsilon M_F}}$		
									$-\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}T_{y_1,a_1,j_2}T_{y_2,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3M_F}}$		
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}I_{12,j_2,j_3,j_1,j_4}^\dagger\lambda_{12}^{(1)}$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}I_{12,j_2,j_3,j_1,j_4}^\dagger\lambda_{12}^{(1)}$	$+3h\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger$ $+\frac{h\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4MF}$ $-\frac{6h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{\epsilon MF}$ $-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{MF}$ $+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{MF}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4\epsilon MF}$ $-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{8MF}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{2\epsilon MF}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4MF}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4\epsilon MF}$ $-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{8MF}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4\epsilon MF}$ $-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4\epsilon MF}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{8MF}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{2\epsilon MF}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}T_{y_1,a_1,j_1}T_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4MF}$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4\epsilon MF}$ $+\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4MF}$	$-2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}T_{y_1,a_1,j_2}T_{y_1,a_2,j_1}H_{a_1}^\dagger S_{1,j_1}^\dagger$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger}I_{H,1,a_2,j_1,j_2}^\dagger\lambda_{12}^{(1)}$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger}I_{H,1,a_2,j_1,j_2}^\dagger\lambda_{H1}^{(1)}$	
cont.									$+\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}$ $-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}$ $+\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}$	$-\frac{h\overline{y_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$	
cont.									$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}$		
cont.									$-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}S_{1,j_3}S_{2,j_$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$+\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}{M_F}{}^T C P L_{L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}}{}^\dagger\lambda_{12}^{(1)}$			
													$+\frac{h\overline{L}_{a_1,i_1}}{\epsilon M_F}{}^T C P R_{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}$			
													$+\frac{h\overline{L}_{a_1,i_1}}{M_F}{}^T C P R_{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}$			
cont.													$-\frac{h\overline{y}_{1,i_2,r_1}y_{2,i_1,r_1}L_{a_2,i_2}}{\sqrt{3}\epsilon M_F}{}^T C P L_{L_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}I_{y_2,a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}$			
													$-\frac{h\overline{y}_{1,i_2,r_1}y_{2,i_1,r_1}L_{a_2,i_2}}{\sqrt{3}M_F}{}^T C P L_{L_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}I_{y_2,a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}$			
													$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}{\sqrt{3}\epsilon M_F}{}^T C P L_{L_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}I_{y_2,a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}$			
cont.													$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}{\sqrt{3}M_F}{}^T C P L_{L_{a_1,i_1}S_{2,j_2}I_{y_1,a_1,j_1}I_{y_2,a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}$			
													$-\frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}\epsilon M_F}{}^T C P R_{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}$			
													$-\frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}{}^T C P R_{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}$			
cont.													$-\frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}\epsilon M_F}{}^T C P R_{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}$			
													$-\frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}{}^T C P R_{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}$			
													$-\frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}\epsilon M_F}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}$			
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$						$+\frac{2}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$			
													$-\frac{10}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$			
													$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y}_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger I_{x_{12},j_2,j_3,j_1,j_4}}{}^1\lambda_{12}^{(1)}$			
cont.													$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y}_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger I_{x_{12},j_2,j_3,j_1,j_4}}{}^1\lambda_{12}^{(1)}$			
													$+\frac{3h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y}_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$			
													$-\frac{3h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y}_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$			
cont.													$+\frac{3h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y}_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$			
													$-\frac{3h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\overline{y}_{2,i_1,r_1}}S_{1,j$			

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+C_{\frac{1}{2}g_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{2,1,1} \overline{r}_1 S_{1,j_1} S_{2,j_2} y_{1,1,1} \overline{r}_1 \text{Bar}(\text{eg3})_{j_1,j_2}}$ $- \frac{5}{9} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{2,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_2 S_{1,j_1} S_{1,j_3} S_{2,j_2} S_{2,j_4} y_{1,1,1} \overline{r}_2 y_{1,1,2} \overline{r}_1 \text{Bar}(\text{eg3})_{j_1,j_2} \text{Bar}(\text{eg3})_{j_3,j_4}}$ $+ \frac{10h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_2 S_{1,j_1} S_{2,j_2} y_{1,1,1} \overline{r}_2 y_{1,1,2} \overline{r}_1 \text{Bar}(\text{eg3})_{j_2,j_3} S_{2,j_1}^\dagger}}{3\sqrt{3}}$ $+ \frac{10h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{2,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_2 S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,1,1} \overline{r}_2 y_{2,1,2} \overline{r}_1 \text{Bar}(\text{eg3})_{j_2,j_3} S_{2,j_1}^\dagger}}{3\sqrt{3}}$	$- \frac{3h \overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,j_2} \overline{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger g_2^2}{8M_F}$ $- \frac{h \overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,j_2} \overline{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger g_2^2}{8M_F}$ $+ \frac{9\sqrt{3}h \overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger g_2^2}{8M_F}$ $+ \frac{3\sqrt{3}h \overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger g_2^2}{8M_F}$	$-2\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{2,1,1} \overline{r}_1 H_{a_1} H_{a_2} S_{1,j_1} S_{1,j_2} y_{1,1,1} \overline{r}_1 \lambda_{T3} \text{Bar}(\text{cg1})_{a_1,j_1,a_2,j_2}}$ $- \frac{4h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 Y_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$ $-2\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 \lambda_{T3} \text{cg1}_{a_1,j_1,a_2,j_2} y_{2,1,1} \overline{r}_1 H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger S_{1,j_2}^\dagger}$ $+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,j_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}^\dagger}}{\sqrt{3}}$
cont.								$+C_{\frac{1}{2}g_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 \text{cg3}_{j_1,j_2} y_{2,1,1} \overline{r}_1 S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+ \frac{2}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_2 S_{1,j_2} S_{2,j_1} y_{1,1,1} \overline{r}_2 y_{1,1,2} \overline{r}_1 S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $- \frac{10}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_2 S_{1,j_1} S_{2,j_2} y_{1,1,1} \overline{r}_2 y_{2,1,2} \overline{r}_1 S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+ \frac{8}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_2 S_{1,j_2} S_{2,j_1} y_{1,1,1} \overline{r}_2 y_{2,1,2} \overline{r}_1 S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $- \frac{10}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_2 S_{1,j_1} S_{2,j_2} y_{1,1,1} \overline{r}_2 y_{2,1,2} \overline{r}_1 S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$	$- \frac{3h L_{a_1,1} C P_R \overline{L}_{a_2,1,2}{}^T S_{1,j_2} y_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_1 \overline{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger g_2^2}{8M_F}$ $- \frac{h L_{a_1,1} C P_R \overline{L}_{a_2,1,2}{}^T S_{1,j_2} y_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_1 \overline{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger g_2^2}{8M_F}$ $+ \frac{9\sqrt{3}h L_{a_1,1} C P_R \overline{L}_{a_2,1,2}{}^T S_{1,j_1} y_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_1 \epsilon_{a_1,a_2} S_{2,j_1}^\dagger g_2^2}{8M_F}$ $+ \frac{3\sqrt{3}h L_{a_1,1} C P_R \overline{L}_{a_2,1,2}{}^T S_{1,j_1} y_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_1 \epsilon_{a_1,a_2} S_{2,j_1}^\dagger g_2^2}{8M_F}$	$- \sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 H_{a_2} S_{1,j_2} y_{1,1,1} \overline{r}_1 H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}^\dagger \lambda_{11}^{(1)}}$ $- \sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 H_{a_2} S_{1,j_2} y_{1,1,1} \overline{r}_1 H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}^\dagger 2\lambda_{12}^{(1)}}$
cont.								$+ \frac{10h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 y_{1,i_2} \overline{r}_2 \text{cg3}_{j_1,j_2} S_{1,j_1} y_{1,1,1} \overline{r}_2 y_{2,1,2} \overline{r}_1 S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger}}{3\sqrt{3}}$ $+ \frac{10h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_2 \text{cg3}_{j_1,j_2} S_{2,j_2} y_{1,1,1} \overline{r}_2 y_{2,1,2} \overline{r}_1 S_{1,j_1}^\dagger S_{2,j_3}^\dagger}}{3\sqrt{3}}$	$- \frac{h \overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} g_2^2 S_{2,j_2} \overline{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8M_F}$ $- \frac{\overline{y}_{1,1,1} \overline{r}_1 y_{2,1,2} \overline{r}_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,j_2} \overline{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{2M_F}$ $+ \frac{\sqrt{3}h \overline{y}_{1,1,1} \overline{r}_1 y_{1,i_3} \overline{r}_2 y_{2,1,2} \overline{r}_2 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,j_2}$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\begin{aligned} & +\frac{2}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\frac{10}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_1,j_4}^{-1} \lambda_{12}^{(1)} \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_1,j_4}^{-1} \lambda_{12}^{(1)} \end{aligned}$	$\begin{aligned} & -\frac{3 h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}}{8 e M_F} T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & -\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & -\frac{\overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}}{2 M_F} T C P_L L_{a_1,i_1} S_{2,j_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{\sqrt{3} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}{8 e M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{3 \sqrt{3} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}{16 M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{\sqrt{3} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{4 e M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \end{aligned}$	$\begin{aligned} & -\frac{4 h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} Y_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}} \\ & +\frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} V_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I_{H1,a_1,a_2,j_1,j_2}}^{-1}}{\sqrt{3}} \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I_{H1,a_1,a_2,j_1,j_2}}^{-1} \lambda_{H1}^{(1)} \end{aligned}$
cont.								$\begin{aligned} & +3 h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & -3 h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & +3 h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_2} S_{2,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \\ & -3 h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_1} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)} \end{aligned}$	$\begin{aligned} & +\frac{3 \sqrt{3} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}{16 M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{\sqrt{3} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{4 e M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{\sqrt{3} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{\sqrt{3} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \end{aligned}$	$-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I_{H1,a_1,a_2,j_1,j_2}}^{-2} \lambda_{H1}^{(2)}$
cont.								$\begin{aligned} & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_1,j_4}^{-2} \lambda_{12}^{(2)} \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_1,j_4}^{-2} \lambda_{12}^{(2)} \end{aligned}$	$\begin{aligned} & +\frac{3 \sqrt{3} h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{16 M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_2,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{\sqrt{3} h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 e M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{3 \sqrt{3} h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{16 M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{\sqrt{3} h \overline{y_{1,i_3,r_1}} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{4 e M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \end{aligned}$	
cont.								$\begin{aligned} & +\frac{\sqrt{3} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{9 \sqrt{3} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \\ & +\frac{9 \sqrt{3} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \end{aligned}$		
cont.								$\begin{aligned} & +\frac{3 \sqrt{3} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{3 \sqrt{3} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \end{aligned}$		
cont.								$\begin{aligned} & -\frac{\sqrt{3} \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{3 h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}{8 e M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{9 h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}{16 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{3 h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{4 e M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{3 h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \end{aligned}$		
cont.								$\begin{aligned} & +\frac{3 h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{8 e M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{9 h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{16 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{3 h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{8 e M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{9 h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}{16 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \end{aligned}$		
cont.								$\begin{aligned} & +\frac{3 h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{8 e M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{9 h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}{16 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{3 h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}}{8 e M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{9 h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}}{16 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \end{aligned}$		
cont.								$\begin{aligned} & +\frac{3 h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}}{8 e M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{9 h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}}{16 M_F} T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \\ & +\frac{\sqrt{3} h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{8 e M_F} \\ & +\frac{3 \sqrt{3} h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{16 M_F} \end{aligned}$		
cont.								$\begin{aligned} & -\frac{3 h L_{a_1,i_1} C P_R L_{a_2,i_2}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{8 e M_F} \\ & -\frac{h L_{a_1,i_1} C P_R L_{a_2,i_2}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{8 M_F} \\ & -\frac{\overline{L_{a_1,i_1}} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{2 M_F} \\ & +\frac{\sqrt{3} h \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{8 e M_F} \\ & +\frac{3 \sqrt{3} h \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{16 M_F} \\ & +\frac{\sqrt{3} h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{1,i_3,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{4 e M_F} \\ & +\frac{\sqrt{3} h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{1,i_3,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{8 M_F} \\ & +\frac{\sqrt{3} h \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{4 e M_F} \end{aligned}$		
cont.								$\begin{aligned} & +\frac{\sqrt{3} h \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{8 M_F} \\ & +\frac{3 \sqrt{3} h \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{16 M_F} \\ & +\frac{\sqrt{3} h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{4 e M_F} \\ & +\frac{\sqrt{3} h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{8 M_F} \end{aligned}$		
cont.								$\begin{aligned} & +\frac{\sqrt{3} h \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{8 M_F} \\ & +\frac{3 \sqrt{3} h \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{16 M_F} \\ & +\frac{\sqrt{3} h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{4 e M_F} \\ & +\frac{\sqrt{3} h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{2,j_1}^\dagger}{8 M_F} \end{aligned}$		

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
													$-\frac{\sqrt{3L_{a_1, i_1}} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{2M_F}$			
cont.													$-\frac{3h\overline{y_{1, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_3, r_2} y_{2, i_1, r_2} y_{2, i_2, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{8\epsilon M_F}$ $-\frac{9h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_3, r_2} y_{2, i_1, r_2} y_{2, i_2, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{16M_F}$ $+\frac{3h\overline{y_{1, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_4, r_2} y_{1, i_3, r_1} y_{2, i_2, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4\epsilon M_F}$			
cont.													$+\frac{3h\overline{y_{1, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_1} y_{2, i_2, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{8M_F}$ $+\frac{3h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{2, i_2, r_2} y_{2, i_3, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4\epsilon M_F}$ $+\frac{3h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{2, i_2, r_2} y_{2, i_3, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{8M_F}$			
cont.													$+\frac{3h\overline{y_{1, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{8\epsilon M_F}$ $+\frac{9h\overline{y_{1, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{16M_F}$ $+\frac{3h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{8\epsilon M_F}$			
cont.													$+\frac{9h\overline{y_{2, i_3, r_1}} L_{a_1, i_1} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{16M_F}$ $+\frac{\sqrt{3}h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$ $+\frac{\sqrt{3}h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$			
cont.													$+\frac{\sqrt{3}h\overline{L_{a_1, i_1}} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$ $+\frac{\sqrt{3}h\overline{L_{a_1, i_1}} C P_R L_{a_2, i_2}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$ $+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$			
cont.													$+\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$ $+\frac{h\overline{L_{a_1, i_1}} C P_R L_{a_2, i_2}^T S_{1, j_2} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T_{3, a_1, a_2, j_1, j_2} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$ $+\frac{h\overline{L_{a_1, i_1}} C P_R L_{a_2, i_2}^T S_{1, j_2} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T_{3, a_1, a_2, j_1, j_2} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$			
T3-C	2	2	1	2	2	3	$\frac{3}{2}$						$+\frac{2}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} y_{2, i_2, r_2} S_{1, j_2} S_{2, j_1} y_{1, i_2, r_1} y_{2, i_1, r_2} S_{1, j_1}^\dagger S_{2, j_2}^\dagger}{M_F^2}$ $-\frac{10}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1, i_1, r_1}} y_{2, i_2, r_2} S_{1, j_1} S_{2, j_2} y_{1, i_2, r_1} y_{2, i_1, r_2} S_{1, j_1}^\dagger S_{2, j_2}^\dagger}{M_F^2}$		$-\frac{4h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1, i_1, r_1}} Y_{e, i_2, i_3} H_{a_1} S_{1, j_1} y_{1, i_2, r_1} Y_{e, i_1, i_3} H_{a_1}^\dagger S_{1, j_1}^\dagger}{\sqrt{3}}$	
													$-\frac{h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{8M_F}$ $+\frac{3h\overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^T C P_L L_{a_1, i_1} g_Y^2 S_{2, j_2} \mathcal{I} T_{3, a_1, a_2, j_2, j_1} S_{1, j_1}^\dagger}{\epsilon M_F}$		$+\frac{2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1, i_1, r_1}} Y_{e, i_2, i_3} H_{a_2} S_{1, j_2} y_{1, i_2, r_1} Y_{e, i_1, i_3} H_{a_1}^\dagger S_{1, j_1}^\dagger \mathcal{I} H_{1, a_1, a_2, j_1, j_2}^{-1}}{\sqrt{3}}$	
cont.													$+\frac{h\overline{y_{1, i_1,$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
													$-\frac{\sqrt{3}\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{2M_F}$			
cont.													$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8\epsilon M_F}$			
													$+\frac{9h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{16M_F}$			
													$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$			
cont.													$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8M_F}$			
													$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8\epsilon M_F}$			
													$+\frac{9h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{16M_F}$			
cont.													$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8\epsilon M_F}$			
													$+\frac{9h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{16M_F}$			
													$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8\epsilon M_F}$			
cont.													$+\frac{9h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{16M_F}$			
													$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_2}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$			
													$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_2}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8\epsilon M_F}$			
cont.													$+\frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_2}}\overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8\epsilon M_F}$			
													$+\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{16M_F}$			
													$+\frac{3h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T \tilde{a}_{a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{8\epsilon M_F}$			
cont.													$-\frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T \tilde{a}_{a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{8M_F}$			
													$+\frac{3h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_Y^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T \tilde{a}_{a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{\epsilon M_F}$			
													$+\frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_Y^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T \tilde{a}_{a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{M_F}$			
cont.													$-\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T \tilde{a}_{a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{2M_F}$			
													$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
									$+ \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
cont.									$+ \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+ \frac{9h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+ \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
cont.									$+ \frac{9h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+ \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+ \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.									$+ \frac{\sqrt{3}h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+ \frac{\sqrt{3}h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$+ \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} \overline{\mathcal{I}T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$+ \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} \overline{\mathcal{I}T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$+ \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$+ \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$- \frac{4h \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_2}} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$- \frac{2h \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}2_{12,j_2,j_1}} \lambda_{12}^{(1)}}{\sqrt{3}}$	$- \frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	$- \frac{2h \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$
								$- \frac{2h \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}2_{12,j_2,j_1}} \lambda_{12}^{(1)}}{\sqrt{3}}$	$+ \frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger}{M_F}$	$- 2i\sqrt{\frac{2}{3}} h \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}H_{1,a_1,a_2,j_2,j_1}} \lambda_{12}^{(1)}$
								$- 2h \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_2 S_{1,j_2} y_{2,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}2_{12,j_2,j_1}} \lambda_{12}^{(1)}}{\sqrt{3}}$	$- \frac{2h \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{\mathcal{I}H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}H_{1,a_1,a_2,j_2,j_1}}} \lambda_{12}^{(2)}}{\sqrt{3}}$	
cont.									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{\sqrt{3}\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{\sqrt{3}M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
cont.									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{2M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
cont.									$- \frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{4M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
cont.									$+ \frac{\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{M_F}$	
									$+ \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$+ \frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
									$- \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{\sqrt{3}\epsilon M_F}$	
									$- \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{\sqrt{3}M_F}$	
									$- \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$- \frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
									$- \frac{h\overline{y_{1,i_3,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$- \frac{3h\overline{y_{1,i_3,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
cont.									$- \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$- \frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
									$+ \frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
									$+ \frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{2M_F}$	
cont.									$- \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{4h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_2^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$	$-\frac{\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1} S_2^\dagger} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$
								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger I_{12,j_2,j_1}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	$-2i\sqrt{\frac{2}{3}} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_2,j_1}^{(2)}}$
								$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_2 S_{1,j_2} y_{2,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger I_{12,j_2,j_1}^1 \lambda_{12}^{(1)}}$	$+\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_2,j_1}^{(2)} \lambda_{12}^{(2)}}{\sqrt{3}}$
cont.								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{\sqrt{3}\epsilon M_F}$	$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{2\sqrt{3}M_F}$	
cont.								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{\epsilon M_F}$	$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{2M_F}$	
cont.								$-\frac{\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	
cont.								$-\frac{3\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	
cont.								$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	$+\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{M_F}$	
cont.								$-\frac{\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{\sqrt{3}\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	
cont.								$-\frac{\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
cont.								$-\frac{3\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	
cont.								$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{2M_F}$	$-\frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger} \lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.								$-\frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger} \lambda_{12}^{(1)}}{\epsilon M_F}$	$-\frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger} \lambda_{12}^{(1)}}{M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger} \lambda_{12}^{(1)}}{\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 \overline{I_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger} \lambda_{12}^{(1)}}{M_F}$	
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{4h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_2^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$	$-\frac{\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1} S_2^\dagger} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$
								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger I_{12,j_2,j_1}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	$-2i\sqrt{\frac{2}{3}} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_2,j_1}^{(2)}}$
								$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_2 S_{1,j_2} y_{2,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger I_{12,j_2,j_1}^1 \lambda_{12}^{(1)}}$	$-\frac{9h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{\tilde{g}}^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_2,j_1}^{(2)} \lambda_{12}^{(2)}}{\sqrt{3}}$
cont.								$-\frac{3h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{\tilde{g}}^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{4M_F}$	$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	
cont.								$+\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{M_F}$	$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{\sqrt{3}\epsilon M_F}$	$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{I_{T3,a_1,a_2,j_1} S_2^\dagger}}{2\sqrt{3}M_F}$	

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$-\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$	
									$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	
cont.									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{11}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
cont.									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
cont.									$-\frac{7h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
cont.									$-\frac{7h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$-\frac{\sqrt{6}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger g_2^2}{\epsilon M_F}$	$-\frac{2h\log\left(\frac{\hat{a}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{\overline{e},i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$
								$+\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$-\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger g_2^2}{M_F}$	$-2i\left(\sqrt{\frac{\hat{\rho}^2}{M_F^2}}\right)\overline{y}_{1,i_1,r_1}Y_{\overline{e},i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger$
								$-\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$-\frac{2\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger g_2^2}{\epsilon M_F}$	$-\frac{2h\log\left(\frac{\hat{a}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_3,j_1,j_4,j_2}^\dagger\lambda_{12}^{(1)}$	$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger g_2^2}{\sqrt{3}M_F}$	
								$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_3,j_1,j_4,j_2}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{\sqrt{6}h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger g_2^2}{\epsilon M_F}$	
								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$-\frac{\sqrt{\frac{2}{3}}h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger g_2^2}{M_F}$	
cont.								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$-\frac{2\sqrt{3}h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger g_2^2}{\epsilon M_F}$	
								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$-\frac{2h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger g_2^2}{\sqrt{3}M_F}$	
								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\epsilon M_F}$	
cont.								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$+\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{6}M_F}$	
								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$	
								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
													$+ \frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$+ \frac{\sqrt{3}h\overline{L}_{a_1,i_1}}{4\epsilon M_F}CPR\overline{L}_{a_2,i_2}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$+ \frac{h\overline{L}_{a_1,i_1}}{4\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$+ \frac{\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$+ \frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$+ \frac{3h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$- \frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{3\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$- \frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{6M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$- \frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{3\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$- \frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{6M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$- \frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$- \frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$- \frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
													$- \frac{3h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$			
cont.													$+ \sqrt{\frac{2}{3}}\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\epsilon M_F}L_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$			
													$+ \sqrt{\frac{2}{3}}\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{M_F}L_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\sqrt{3}\epsilon M_F}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\sqrt{3}M_F}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$+ \sqrt{\frac{2}{3}}\frac{h\overline{L}_{a_1,i_1}}{\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$+ \sqrt{\frac{2}{3}}\frac{h\overline{L}_{a_1,i_1}}{M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$			
cont.													$- 3\sqrt{\frac{2}{11}}\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\epsilon M_F}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$			
													$- 3\sqrt{\frac{2}{11}}\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{M_F}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$			
cont.													$- \frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\sqrt{11}\epsilon M_F}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$			
													$- \frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\sqrt{11}M_F}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$			
cont.													$- 3\sqrt{\frac{2}{11}}\frac{h\overline{L}_{a_1,i_1}}{\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}^\dagger\lambda_{12}^{(2)}$			
													$- 3\sqrt{\frac{2}{11}}\frac{h\overline{L}_{a_1,i_1}}{M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}^\dagger\lambda_{12}^{(2)}$			
cont.													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{11}\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}$			
													$- \frac{h\overline{L}_{a_1,i_1}}{\sqrt{11}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}$			
cont.													$- \frac{7h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\sqrt{23}\epsilon M_F}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}$			
													$- \frac{7h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{\sqrt{23}M_F}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}$			
cont.													$- \frac{7h\overline{L}_{a_1,i_1}}{\sqrt{23}\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}$			
													$- \frac{7h\overline{L}_{a_1,i_1}}{\sqrt{23}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}$			
cont.													$- \frac{7h\overline{L}_{a_1,i_1}}{\sqrt{23}\epsilon M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}$			
													$- \frac{7h\overline{L}_{a_1,i_1}}{\sqrt{23}M_F}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$				$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
											$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
											$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
											$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
											$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
											$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
											$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
											$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
											$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$- \frac{4}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_$				

[illegible]

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$			
													$-3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$ $-\frac{3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$ $-\frac{3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{a}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
cont.													$-3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}$ $-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$ $-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{a}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
													$-7h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}$ $-\frac{7h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$ $-\frac{7h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$			
													$-7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$			
cont.													$-\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$ $+\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $-\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $-2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}\lambda_{12}^{(1)}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}\lambda_{12}^{(1)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}\lambda_{12}^{(1)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}\lambda_{12}^{(2)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}\lambda_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}\lambda_{12}^{(2)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}\lambda_{12}^{(2)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}\lambda_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(2)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(2)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_1,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_1,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_1,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_1,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}\lambda_{12}^{(3)}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}{\sqrt{3}M_F}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
									$-\frac{2\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}{\epsilon M_F}{}^T C P_L L_{a_1,i_1}g_Y^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
									$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}{+\frac{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}{\sqrt{3}M_F}}{}^T C P_L L_{a_1,i_1}g_Y^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
									$-\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{3\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{6M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{+\frac{4\epsilon M_F}{8M_F}}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
									$-\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
cont.									$-\frac{h\overline{y}_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
									$-\frac{3h\overline{y}_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
cont.									$-\frac{3h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}L_{a_2,i_2}}{+\frac{3\epsilon M_F}{6M_F}}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger$	
									$-\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{2\sqrt{2}\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
cont.									$-\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{4\sqrt{2}M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
									$-\frac{\sqrt{6}hL_{a_1,i_1}}{\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T g_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
									$-\frac{\sqrt{\frac{2}{3}}hL_{a_1,i_1}}{M_F}C P_R \overline{L}_{a_2,i_2}{}^T g_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
cont.									$-\frac{2\sqrt{6}hL_{a_1,i_1}}{\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T g_Y^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
									$-\frac{2\sqrt{\frac{2}{3}}hL_{a_1,i_1}}{M_F}C P_R \overline{L}_{a_2,i_2}{}^T g_Y^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
									$+\frac{\sqrt{\frac{2}{3}}L_{a_1,i_1}}{M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
cont.									$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}}{2\sqrt{2}\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
									$-\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}}{4\sqrt{2}M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
									$-\frac{\sqrt{2}h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{3\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_1,r_2}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
cont.									$-\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{3\sqrt{2}M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_1,r_2}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
									$-\frac{\sqrt{2}h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}}{3\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
									$-\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{3\sqrt{2}M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
cont.									$-\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{4\sqrt{2}M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
									$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}}{2\sqrt{2}\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_2}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
									$-\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}}{4\sqrt{2}M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_2}\overline{I}\tau_{3a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}{}^\dagger$	
cont.									$+\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{4\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger$	
									$+\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{8M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger$	
									$-\frac{2\sqrt{3}hL_{a_1,i_1}}{\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger$	
cont.									$-\frac{2hL_{a_1,i_1}}{\sqrt{3}M_F}C P_R \overline{L}_{a_2,i_2}{}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger$	
									$-\frac{2\sqrt{3}hL_{a_1,i_1}}{\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T g_Y^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger$	
									$-\frac{2hL_{a_1,i_1}}{\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T g_Y^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger$	
cont.									$+\frac{L_{a_1,i_1}}{+\frac{L_{a_1,i_1}}{\sqrt{3}M_F}}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger$	
									$+\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}}{4\epsilon M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger$	
									$+\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}}{8M_F}C P_R \overline{L}_{a_2,i_2}{}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{3\epsilon M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_2y_{1,i_3},r_1y_{2,i_2},r_2\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$	
									$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{6M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_2y_{1,i_3},r_1y_{2,i_2},r_2\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$	
									$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{6M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_2y_{2,i_2},r_2y_{2,i_3},r_1\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$	
cont.									$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{6M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_2y_{2,i_2},r_2y_{2,i_3},r_1\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$	
									$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_2y_{1,i_3},r_1y_{2,i_3},r_2\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$	
									$-\frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_2y_{1,i_2},r_1y_{2,i_3},r_2\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$	
cont.									$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_2y_{2,i_3},r_1y_{2,i_3},r_2\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$	
									$-\frac{3h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_2y_{2,i_2},r_1y_{2,i_3},r_2\epsilon_{a_1,a_2}S_{2,j_1}^\dagger$	
									$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1},r_1y_{2,i_2},r_1L_{a_2,i_2}}{\epsilon M_F}C P_L L_{a_1,i_1}S_{2,j_2}\overline{I}T\mathfrak{z}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$	
cont.									$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1},r_1y_{2,i_2},r_1L_{a_2,i_2}}{M_F}C P_L L_{a_1,i_1}S_{2,j_2}\overline{I}T\mathfrak{z}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$	
									$-\frac{h\overline{y}_{1,i_1},r_1y_{2,i_2},r_1L_{a_2,i_2}}{\sqrt{3}\epsilon M_F}C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$	
									$-\frac{h\overline{y}_{1,i_1},r_1y_{2,i_2},r_1L_{a_2,i_2}}{\sqrt{3}M_F}C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}$	
cont.									$+\frac{\sqrt{\frac{2}{3}}h\overline{L}_{a_1,i_1}}{\epsilon M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1},r_1y_{2,i_2},r_1\overline{I}T\mathfrak{z}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}$	
									$+\frac{\sqrt{\frac{2}{3}}h\overline{L}_{a_1,i_1}}{M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1},r_1y_{2,i_2},r_1\overline{I}T\mathfrak{z}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}$	
									$-\frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}\epsilon M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_1y_{2,i_2},r_1\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$	
cont.									$-\frac{h\overline{L}_{a_1,i_1}}{\sqrt{3}M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_1y_{2,i_2},r_1\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1},r_1y_{2,i_2},r_1L_{a_2,i_2}}{\epsilon M_F}C P_L L_{a_1,i_1}S_{2,j_2}\overline{I}T\mathfrak{z}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1},r_1y_{2,i_2},r_1L_{a_2,i_2}}{M_F}C P_L L_{a_1,i_1}S_{2,j_2}\overline{I}T\mathfrak{z}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$	
cont.									$-\frac{h\overline{y}_{1,i_1},r_1y_{2,i_2},r_1L_{a_2,i_2}}{\sqrt{11}\epsilon M_F}C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$	
									$-\frac{h\overline{y}_{1,i_1},r_1y_{2,i_2},r_1L_{a_2,i_2}}{\sqrt{11}M_F}C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{L}_{a_1,i_1}}{\epsilon M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1},r_1y_{2,i_2},r_1\overline{I}T\mathfrak{z}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}$	
cont.									$-\frac{3\sqrt{\frac{2}{11}}h\overline{L}_{a_1,i_1}}{M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2}y_{1,i_1},r_1y_{2,i_2},r_1\overline{I}T\mathfrak{z}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}$	
									$-\frac{h\overline{L}_{a_1,i_1}}{\sqrt{11}\epsilon M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_1y_{2,i_2},r_1\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}$	
									$-\frac{h\overline{L}_{a_1,i_1}}{\sqrt{11}M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_1y_{2,i_2},r_1\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}$	
cont.									$-\frac{7h\overline{y}_{1,i_1},r_1y_{2,i_2},r_1L_{a_2,i_2}}{\sqrt{23}\epsilon M_F}C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}$	
									$-\frac{7h\overline{y}_{1,i_1},r_1y_{2,i_2},r_1L_{a_2,i_2}}{\sqrt{23}M_F}C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}$	
									$-\frac{7h\overline{L}_{a_1,i_1}}{\sqrt{23}\epsilon M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_1y_{2,i_2},r_1\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}$	
cont.									$-\frac{7h\overline{L}_{a_1,i_1}}{\sqrt{23}M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_1y_{2,i_2},r_1\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}$	
									$-\frac{7h\overline{L}_{a_1,i_1}}{\sqrt{23}\epsilon M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_1y_{2,i_2},r_1\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}$	
									$-\frac{7h\overline{L}_{a_1,i_1}}{\sqrt{23}M_F}C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1}y_{1,i_1},r_1y_{2,i_2},r_1\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}$	

Field combinations 8 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F$	ℓ	$S_2 F$	ℓ
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$				
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0				
T3-A	0	1	0	3	1	2	$\frac{1}{2}$				
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1				
T3-A	2	1	1	3	2	2	$\frac{3}{2}$				
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$				
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0				
cont.											
cont.											
cont.											
T3-B	0	2	0	2	1	1	$\frac{1}{2}$				

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F \ell$	$S_2 F H$	$S_2 F \ell$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1		$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y_{2,a_1,j_1}} I_{y_{2,a_2,j_2}} H_{a_1}^\dagger S_{2,j_1}^\dagger$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$		
T3-B	2	2	1	2	2	1	$\frac{3}{2}$		$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y_{2,a_1,j_1}} I_{y_{2,a_2,j_2}} H_{a_1}^\dagger S_{2,j_1}^\dagger$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$		
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$		$-\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}^{-1}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$		
cont.											
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0		$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} c \bar{q}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_1} y_{l,i_2,r_1} y_{l,i_2}^2}{8 M_F}$ $+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} \text{Bar}(c q_{2,a_1,j_1,a_2,j_2}) S_{l,j_1}^\dagger S_{l,j_2}^\dagger g_2^2}{8 M_F}$ $+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{q}_{a_1,j_1,a_2,j_2} g_2^2 S_{i,j_1} S_{i,j_2} y_{l,i_1,r_1} y_{l,i_2,r_1}}{8 M_F}$ $+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{2 M_F}$	$-\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}^{-1}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H2,a_1,a_2,j_1,j_2}^{1\lambda_{H2}^{(1)}}$ $-\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H2,a_1,a_2,j_1,j_2}^{2\lambda_{H2}^{(2)}}$	$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(c q_{1,a_1,j_1,a_2,j_2}) y_{2,i_2}^2}{8 M_F}$ $+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{q}_{1,a_1,j_1,a_2,j_2} y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger g_2^2}{8 M_F}$ $+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_1} S_{2,j_2} \text{Bar}(c q_{1,a_1,j_1,a_2,j_2})}{8 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_2} y_{l,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{l,i_3,r_1} \text{Bar}(c q_{1,a_1,j_1,a_2,j_2})}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_1,r_1}} y_{l,i_2,r_2} y_{l,i_3,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{l,i_3,r_2} \text{Bar}(c q_{1,a_1,j_1,a_2,j_2})}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2} y_{l,i_2,r_2} y_{l,i_3,r_1}}{4 M_F}$ $-\frac{\sqrt{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{l,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}{}^T c \bar{g}_{a_1,j_1,a_2,j_2} S_{i,j_1} S_{i,j_2} y_{l,i_1,r_2$

[illegible]

Field combinations 9 of 10

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.										$\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_1 y_{1, i_3, r_2} y_{2, i_1, r_2} y_{2, i_2, r_1} \overline{x}_{T \beta_1, a_2, j_1} S_{2, j_1}^\dagger}{4 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_1 y_{1, i_1, r_2} y_{1, i_3, r_2} y_{2, i_2, r_2} \overline{x}_{T \beta_1, a_2, j_1} S_{2, j_1}^\dagger}{M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_1 y_{1, i_1, r_2} y_{2, i_2, r_2} y_{2, i_3, r_1} \overline{x}_{T \beta_1, a_2, j_1} S_{2, j_1}^\dagger}{\sqrt{3} M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_1 y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} T \tau_{\beta_1, a_2, j_1} S_{2, j_1}^\dagger}{4 M_F}$
cont.										$\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_1 y_{1, i_1, r_2} y_{2, i_2, r_1} y_{2, i_3, r_2} \overline{x}_{T \beta_1, a_2, j_1} S_{2, j_1}^\dagger}{4 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} \overline{y}_{2, i_2, r_1} L_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_1} \overline{x}_{T \beta_1, a_2, j_1} S_1^\dagger \lambda_{12}^{(1)}}{M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_1 y_{1, i_1, r_1} y_{1, r_2} y_{2, i_2, r_2} \overline{x}_{T \beta_1, a_2, j_1} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} \overline{y}_{2, i_2, r_1} \overline{l}_{a_4, i_2}{}^T C P_L \overline{l}_{a_2, i_1} H_{a_1} H_{a_2} \lambda_{T \beta_3} \epsilon_{a_1, a_2} \epsilon_{a_3, a_4}}{\sqrt{3} M_F}$	$-\frac{1}{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} \overline{\lambda}_{T \beta_3} S_{2, j_2} y_{1, i_1, r_1} \overline{x}_{T \beta_3, a_1, a_2, j_1, j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_2, r_1} \overline{y}_{2, i_1, r_1} \overline{l}_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} g_2^2 S_{2, j_2} \overline{x}_{T \beta_1, a_1, i_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger}{4 M_F}$
								$+\frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda}_{T \beta_3} \overline{l}_{a_3, i_1} C P_R \overline{l}_{a_4, i_2}{}^T y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$-\frac{1}{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2, i_1, r_1} \overline{\lambda}_{T \beta_3} S_{2, j_2} y_{2, i_1, r_1} \overline{x}_{T \beta_3, a_1, a_2, j_1, j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger$	$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2} \epsilon_{a_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger$
									$-\frac{1}{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} H_{a_1} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} \overline{x}_{T \beta_3, a_1, a_2, j_2, j_1} \lambda_{T \beta_3} S_{2, j_1}^\dagger$	$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2} \epsilon_{a_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger$
										$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} y_{1, i_2, r_2} y_{2, i_3, r_1} \overline{l}_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_2} \overline{x}_{\beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger$
cont.									$-\frac{1}{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2, i_1, r_1} H_{a_1} H_{a_2} S_{1, j_2} y_{2, i_1, r_1} \overline{x}_{T \beta_3, a_1, a_2, j_2, j_1} \lambda_{T \beta_3} S_{2, j_1}^\dagger$	$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_3, r_1} y_{2, i_1, r_2} y_{2, i_2, r_2} \epsilon_{a_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{x}_{\beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger$
										$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} \overline{l}_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{x}_{\beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger$
										$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} \overline{l}_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{x}_{\beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger$
										$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} \overline{l}_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} g_2^2 S_{2, j_2} \overline{x}_{T \beta_1, a_1, i_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger}{4 M_F}$
cont.										$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T g_2^2 S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$
										$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_3, r_2} y_{1, i_2, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger$
										$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_3, r_2} y_{1, i_2, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger$
										$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_3, r_2} y_{1, i_2, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_3, r_2} y_{1, i_2, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger}{4 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_3, r_2} y_{1, i_2, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger}{4 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_3, r_1} \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_3, r_2} y_{1, i_2, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger}{4 M_F}$
										$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T g_2^2 S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$
cont.										$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} \overline{y}_{2, i_2, r_1} L_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_2, r_1} \overline{y}_{2, i_1, r_1} L_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} \overline{y}_{2, i_2, r_1} L_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} \overline{y}_{2, i_2, r_1} \overline{l}_{a_4, i_2}{}^T C P_L \overline{l}_{a_2, i_1} H_{a_1} H_{a_2} \lambda_{T \beta_3} \epsilon_{a_1, a_2} \epsilon_{a_3, a_4}}{\sqrt{3} M_F}$	$-2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2, i_1, r_1} \overline{y}_{2, i_2, r_1} H_{a_2} S_{1, j_1} S_{2, j_2} y_{1, i_2, r_1} y_{1, i_3, r_2} \overline{x}_{T \beta_1, a_2, j_1} \overline{x}_{a_2, j_2} H_{a_1}^\dagger$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_2, r_1} \overline{y}_{2, i_1, r_1} L_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger g_2^2}{4 M_F}$
								$+\frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda}_{T \beta_3} \overline{l}_{a_3, i_1} C P_R \overline{l}_{a_4, i_2}{}^T y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$-\frac{1}{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} \overline{\lambda}_{T \beta_3} S_{2, j_2} y_{2, i_1, r_1} \overline{x}_{T \beta_3, a_1, a_2, j_1, j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} \overline{y}_{2, i_2, r_1} L_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{4 M_F}$
									$-\frac{1}{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2, i_1, r_1} \overline{\lambda}_{T \beta_3} S_{2, j_2} y_{2, i_1, r_1} \overline{x}_{T \beta_3, a_1, a_2, j_1, j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger$	$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
									$-\frac{1}{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} H_{a_1} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} \overline{x}_{T \beta_3, a_1, a_2, j_2, j_1} \lambda_{T \beta_3} S_{2, j_1}^\dagger$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_2, r_1} \overline{y}_{2, i_1, r_1} L_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
cont.									$-\frac{1}{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2, i_1, r_1} H_{a_1} H_{a_2} S_{1, j_2} y_{2, i_1, r_1} \overline{x}_{T \beta_3, a_1, a_2, j_2, j_1} \lambda_{T \beta_3} S_{2, j_1}^\dagger$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger g_2^2}{4 M_F}$
									$-\frac{1}{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} H_{a_1} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} \overline{x}_{T \beta_3, a_1, a_2, j_2, j_1} \lambda_{T \beta_3} S_{2, j_1}^\dagger$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger g_2^2}{4 M_F}$
									$-\frac{1}{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2, i_1, r_1} H_{a_1} H_{a_2} S_{1, j_2} y_{2, i_1, r_1} \overline{x}_{T \beta_3, a_1, a_2, j_2, j_1} \lambda_{T \beta_3} S_{2, j_1}^\dagger$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{l}_{a_1, i_1} C P_R \overline{l}_{a_2, i_2}{}^T S_{1, j_2} y_{1, i_2, r_1} y_{2, i_1, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger g_2^2}{4 M_F}$
									$-2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} \overline{y}_{2, i_2, r_1} H_{a_2} y_{2, j_2, r_1} y_{1, i_3, r_2} L_{a_2, i_2}{}^T S_{2, j_2}{}^\dagger H_{a_1}^\dagger S_{1, j_1}^\dagger S_{2, j_2}{}^\dagger$	$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_2} L_{a_2, i_2}{}^T C P_L \overline{l}_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_1} \overline{x}_{T \beta_1, a_1, j_1} \overline{x}_{a_2, j_2} S_{1, j_1}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$	
cont.									$+h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} H_{a_1}^\dagger \lambda_{H2}^{(1)}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{2M_F}$	
									$+h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{1,a_2,j_2}}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(1)}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger}{4M_F}$	
									$+\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_1} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{M_F}$	
									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_1} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{4M_F}$	
cont.									$+h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg}1_{j_1,j_2}} H_{a_1} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{4M_F}$	
									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1,j_1}} \mathcal{I}_{y_{1,a_2,j_2}}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{2M_F}$	
										$+\frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}^T} g_1^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,j_1}^\dagger}{4M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$	
cont.										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$	
cont.										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	
										$+\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} e_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
										$+\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{a_1,a_2} S_{2,i_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_2,r_1} y_{2,i_1,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
cont.										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} e_{a_1,a_2} e_{a_3,a_4}}{\sqrt{3} M_F}$	$-\frac{1}{2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T3}} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3} a_{a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$+\frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_2,r_1} y_{2,i_1,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} g_2^T S_{2,j_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger}{4M_F}$	
								$+\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1} C P_R L_{a_4,i_2}^T} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{a_1,a_3} e_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$+\frac{1}{2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T3}} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3} a_{a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{4M_F}$	
								$-\frac{1}{2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_1} y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,j_1}^\dagger}{2M_F}$		
									$-\frac{1}{2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{4M_F}$	
cont.										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{4M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{4M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{2M_F}$	
										$+\frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} g_2^T S_{2,j_1} e_{a_1,a_2} S_{1,i_1}^\dagger}{4M_F}$	
cont.										$+\frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}^T} g_2^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{1,i_1}^\dagger}{4M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$	
cont.										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{2M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{4M_F}$	
										$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{y_{1,a_1,j_1}} \mathcal{I}_{y_{2,a_2,j_2}} S_{2,i_1}^\dagger}{M_F}$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,1,2} \tau_2 y_{2,1,3} \tau_1} L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,2} y_{2,1,3} \tau_2 \mathcal{I}_{y_{1,1},1}^* \mathcal{I}_{y_{2,2,2},2} S_{1,1,1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,1,2} \tau_1 y_{2,1,3} \tau_2} L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,2} y_{2,1,3} \tau_2 \mathcal{I}_{y_{1,1},1}^* \mathcal{I}_{y_{2,2,2},2} S_{1,1,1}^\dagger}{2 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,1,2} \tau_1} L_{a_2,1,2}{}^T C P_L L_{a_1,1} g_2^2 S_{2,1,1} \epsilon_{a_1,2} S_{1,1,1}^\dagger}{4 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) L_{a_1,1} C P_R \overline{L_{a_2,1,2}} g_2^2 S_{1,1,2} y_{1,3} \tau_1 y_{2,1,1} \tau_1 \mathcal{I}_{y_{1,1},1} \mathcal{I}_{y_{2,2,2},2} S_{2,1,1}^\dagger}{4 M_F}$
									$+\frac{6 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,1}} C P_R \overline{L_{a_2,1,2}} g_1^2 S_{1,1,2} y_{1,1} \tau_1 y_{2,1,2} \tau_1 \mathcal{I}_{y_{1,1},1} y_{1,2} \mathcal{I}_{y_{2,2,2},2} S_{2,1,1}^\dagger}{M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,3} \tau_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,1,2}} S_{1,1,2} y_{1,1} \tau_2 y_{1,1,3} \tau_1 y_{2,1,2} \tau_2 \mathcal{I}_{y_{1,1},1} \mathcal{I}_{y_{2,2,2},2} S_{2,1,1}^\dagger}{2 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,3} \tau_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,1,2}} S_{1,1,2} y_{1,1} \tau_1 y_{1,1,3} \tau_2 y_{2,1,2} \tau_2 \mathcal{I}_{y_{1,1},1} \mathcal{I}_{y_{2,2,2},2} S_{2,1,1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,1,3} \tau_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,1,2}} S_{1,1,2} y_{1,3} \tau_2 y_{2,1,1} \tau_1 y_{2,1,2} \tau_2 \mathcal{I}_{y_{1,1},1} \mathcal{I}_{y_{2,2,2},2} S_{2,1,1}^\dagger}{4 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,1,3} \tau_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,1,2}} S_{1,1,2} y_{1,1} \tau_2 y_{2,1,2} \tau_2 y_{2,1,3} \tau_1 \mathcal{I}_{y_{1,1},1} \mathcal{I}_{y_{2,2,2},2} S_{2,1,1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,1,3} \tau_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,1,2}} S_{1,1,2} y_{1,1} \tau_2 y_{1,1,2} \tau_1 y_{2,1,2} \tau_2 \mathcal{I}_{y_{1,1},1} \mathcal{I}_{y_{2,2,2},2} S_{2,1,1}^\dagger}{2 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,1,3} \tau_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,1,2}} S_{1,1,2} y_{1,1} \tau_2 y_{1,1,2} \tau_1 y_{2,1,2} \tau_2 \mathcal{I}_{y_{1,1},1} \mathcal{I}_{y_{2,2,2},2} S_{2,1,1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,1,3} \tau_1} \overline{L_{a_1,1}} C P_R \overline{L_{a_2,1,2}} S_{1,1,2} y_{1,1} \tau_2 y_{2,1,2} \tau_1 y_{2,1,3} \tau_2 \mathcal{I}_{y_{1,1},1} \mathcal{I}_{y_{2,2,2},2} S_{2,1,1}^\dagger}{4 M_F}$
cont.									$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,1,2} \tau_1} L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} \epsilon_{a_1,2} S_{1,1,1}^\dagger \lambda_{12}^{(1)}}{M_F}$	$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,1}} C P_R \overline{L_{a_2,1,2}} S_{1,1,1} y_{1,1} \tau_1 y_{2,1,2} \tau_1 \epsilon_{a_1,2} S_{2,1,1}^\dagger \lambda_{12}^{(1)}}{M_F}$
									$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,2} \tau_1 y_{2,1,1} \tau_1} L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,2} \mathcal{I}_{y_{1,1},1}^* \mathcal{I}_{y_{2,2,2},2} S_{1,1,1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,1,2} \tau_1} L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,2} \mathcal{I}_{y_{1,1},1}^* \mathcal{I}_{y_{2,2,2},2} S_{1,1,1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,1,2} \tau_1} L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,2} \mathcal{I}_{y_{1,1},1}^* \mathcal{I}_{y_{2,2,2},2} S_{1,1,1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,1}} C P_R \overline{L_{a_2,1,2}} S_{1,1,2} y_{1,1} \tau_2 y_{1,1,2} \tau_1 y_{2,1,2} \tau_2 \mathcal{I}_{y_{1,1},1} \mathcal{I}_{y_{2,2,2},2} S_{2,1,1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
T3-C	-2	2	-1	2	0	3	-1/2	$\frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,1,2} \tau_1} L_{a_4,1,2}{}^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,2} \epsilon_{a_3,2} a_4}{3 M_F}$ $-\frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,1}} C P_R \overline{L_{a_4,1,2}} y_{1,1,1} \tau_1 y_{2,1,2} \tau_1 \epsilon_{a_1,2} a_2 a_4 H_{a_1}^\dagger H_{a_3}^\dagger}{3 M_F}$	$-\frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 \lambda_{T3} S_{2,1,2} y_{1,1,1} \tau_1 \mathcal{I}_{T3} a_{1,1} a_{2,1,1} S_{2,1,1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,1,1}^\dagger}$ $-\frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,1,1} \tau_1 \lambda_{T3} S_{2,1,2} y_{2,1,1} \tau_1 \mathcal{I}_{T3} a_{1,1} a_{2,1,1} S_{2,1,1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1,1,1}^\dagger}$ $-\frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} H_{a_1} H_{a_2} S_{1,1,2} y_{1,1,1} \tau_1 \mathcal{I}_{T3,1} a_{1,2} y_{2,1,1} \lambda_{T3} S_{2,1,1}^\dagger$	$-\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,1,2} \tau_1} L_{a_2,1,2}{}^T C P_L L_{a_1,1} g_2^2 S_{2,1,1} \mathcal{I}_{T3,1} a_{1,2} y_{2,1,1} S_{1,1,1}^\dagger}{8 M_F}$ $+\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{1,1,2} \tau_2 y_{2,1,2} \tau_2} L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,2} y_{1,1,3} \tau_1 \mathcal{I}_{T3,1} a_{1,2} y_{2,1,1} S_{1,1,1}^\dagger$ $+\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,2} \tau_2 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,2} y_{1,1,3} \tau_2 \mathcal{I}_{T3,1} a_{1,2} y_{2,1,1} S_{1,1,1}^\dagger$ $+\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{1,1,2} \tau_2 y_{2,1,3} \tau_1} L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,2} y_{1,1,3} \tau_2 \mathcal{I}_{T3,1} a_{1,2} y_{2,1,1} S_{1,1,1}^\dagger$ $+\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{2,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,2} y_{1,1,3} \tau_2 \mathcal{I}_{T3,1} a_{1,2} y_{2,1,1} S_{1,1,1}^\dagger$ $+\frac{9 \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,1,2} \tau_1} L_{a_2,1,2}{}^T C P_L L_{a_1,1} g_2^2 S_{2,1,1} \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,2} \tau_2 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_1 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,2} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{4 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1} S_{2,1,1} y_{1,1,3} \tau_2 \epsilon_{a_1,2} S_{1,1,1}^\dagger}{8 M_F}$ $+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1} y_{1,1,2} \tau_2 y_{2,1,3} \tau_1 L_{a_2,1,2}{}^T C P_L L_{a_1,1$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1	S_2	F	H	ℓ
cont.															$-\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$ $+\frac{9\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^\dagger g_2^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$ $-\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$ $-\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4M_F}$ $+\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4M_F}$ $+\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$ $+\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$
	cont.														$+\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4M_F}$ $+\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$ $+\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$ $+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$ $+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_1,j_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0								$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^TC\,P_L\overline{L_{a_2,i_1}}H_{a_1}H_{a_3}\lambda_{T\mathcal{R}}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$ $-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T\mathcal{R}}L_{a_3,i_1}}C\,P_R\overline{L_{a_4,i_2}}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$
	cont.														$-2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}\text{Bar}(\text{eg4})_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger$ $-\frac{1}{2}\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_{T\mathcal{R}}S_{2,j_2}}y_{1,i_1,r_1}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{1}{2}\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_{T\mathcal{R}}S_{2,j_2}}y_{2,i_1,r_1}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{1}{2}\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}\lambda_{T\mathcal{R}}S_{2,j_1}^\dagger$ $-\frac{1}{2}\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}\lambda_{T\mathcal{R}}S_{2,j_1}^\dagger$ $-2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}\text{eg4}}_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $-h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(\text{eg3})_{j_1,j_2}H_{a_1}^\dagger\lambda_{H2}^{(1)}$ $-3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\text{Bar}(\text{eg4})_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger\lambda_{H2}^{(1)}$ $-h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\text{eg3}}_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H2}^{(1)}$ $-3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\text{eg4}}_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H2}^{(1)}$ $+\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\text{Bar}(\text{eg4})_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger\lambda_{H2}^{(2)}$ $+\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\text{eg4}}_{a_1,j_1,a_2,j_2}H_{a_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{H2}^{(2)}$
cont.															$-\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger g_2^T}{8M_F}$ $+\frac{9\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger g_2^T}{8M_F}$ $-\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_1,j_2}S_{2,j_1}^\dagger g_2^T}{8M_F}$ $+\frac{9\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger g_2^T}{8M_F}$ $-\frac{3h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_1}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_1}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{4M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TC\,P_L\overline{L_{a_1,i_1}}S_{2,j_2}y_{1,i_3,r_2}\overline{\mathcal{I}T\mathcal{R}}_{a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$ $+\frac{\sqrt{3}h\log\$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.										$+ \frac{3h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}}^{\dagger}}{4M_F}$ $+ \frac{3h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}}^{\dagger}}{4M_F}$ $+ \frac{3h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}}^{\dagger}}{8M_F}$ $+ \frac{3h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}}^{\dagger}}{8M_F}$
										$+ \frac{\sqrt{3h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}}^{\dagger} \lambda_{12}^{(1)}}{M_F}$ $+ \frac{\sqrt{3h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}}^{\dagger} \lambda_{12}^{(1)}}{M_F}$ $+ \frac{h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}}^{\dagger} \lambda_{12}^{(2)}}{\sqrt{3}M_F}$ $+ \frac{h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{2,j_1}}^{\dagger} \lambda_{12}^{(2)}}{\sqrt{3}M_F}$
cont.										
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1			$- \frac{2h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}^T} C P_L L_{a_2,i_1} H_{a_1} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3M_F}$ $- \frac{2h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}^T} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}}^{\dagger} H_{a_2}}{3M_F}$
										$- \frac{1}{2} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}}^{\dagger} H_{a_2}}^{\dagger} S_{1,j_1}}^{\dagger}$ $- \frac{1}{2} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}}^{\dagger} H_{a_2}}^{\dagger} S_{1,j_1}}^{\dagger}$ $- \frac{1}{2} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}}^{\dagger}}$
										$- \frac{3h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} g_{S_2}^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}}^{\dagger}}{8M_F}$ $+ \frac{9h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} g_{S_2}^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}}^{\dagger}}{8M_F}$ $+ \frac{\sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}}^{\dagger}}{4M_F}$ $+ \frac{\sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}}^{\dagger}}{4M_F}$ $+ \frac{\sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}}^{\dagger}}{8M_F}$ $+ \frac{\sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}}^{\dagger}}{8M_F}$ $+ \frac{\sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}}^{\dagger}}{8M_F}$ $+ \frac{9\sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} g_{S_2}^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}}^{\dagger}}{8M_F}$ $+ \frac{9\sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} g_{S_2}^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}}^{\dagger}}{8M_F}$
cont.										
cont.										$+ \frac{3h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}}^{\dagger}}{8M_F}$ $+ \frac{3h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}}^{\dagger}}{4M_F}$ $+ \frac{3h \log \left(\frac{\hat$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.										$-\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
cont.										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{I T 3}_{a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$
										$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I T 3}_{a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \overline{I T 3}_{a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \overline{I T 3}_{a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$
									$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3}} S_{2,y_1,i_1,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_2,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3}} S_{2,y_2,i_1,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{M_F}$
										$-\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$
cont.										$-\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,y_1,i_3,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{1,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,y_1,i_3,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{1,j_1}^\dagger}{4M_F}$
										$-\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,y_2,i_3,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{1,j_1}^\dagger}{4M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,y_2,i_3,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{1,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,y_2,i_3,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{1,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,y_2,i_3,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{1,j_1}^\dagger}{4M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,y_2,i_3,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{1,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,y_2,i_3,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{1,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,y_2,i_3,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{1,j_1}^\dagger}{4M_F}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \overline{I T 3}_{a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$	$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger g_1^2}{4M_F}$
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \overline{I T 3}_{a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,y_2,i_3,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{1,j_1}^\dagger g_1^2}{4M_F}$
									$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3}} S_{2,y_1,i_1,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$-\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I T 3}_{a_1,a_2,j_1}^\dagger S_{2,j_1}^\dagger}{4M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$	
cont.									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{\sqrt{3} M_F}$	$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{1, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	
									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1} y_{2, i_1, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	
									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger \lambda_{12}^2}{M_F}$	
cont.									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$		
	T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_4, i_2}}^T C P_L L_{a_2, i_1} H_{a_1} H_{a_3} \lambda T_{3, a_1, a_2, a_3, a_4}}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} H_{a_1} H_{a_2} S_{1, j_1} y_{1, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} \lambda T_{3, a_1, a_2, j_1}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_2, r_1} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$
									$+\frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda T_{3, a_3, i_1}} C P_R \overline{L_{a_4, i_2}}^T y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} H_{a_1} H_{a_2} S_{1, j_1} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} \lambda T_{3, a_1, a_2, j_1}^\dagger$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$
$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{\lambda T_{3, a_1, i_1}} S_2 y_{1, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger}{\sqrt{3}}$									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{\sqrt{3} M_F}$		
cont.									$-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} \overline{\lambda T_{3, a_1, i_1}} S_2 y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger$	$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$	
									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{\sqrt{3} M_F}$	
									$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	
cont.									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{\sqrt{3} M_F}$	
									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1} y_{2, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1} y_{2, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	
									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1} y_{2, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1} y_{2, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{M_F}$	
cont.									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1} y_{2, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1} y_{2, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
	T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+\frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1} y_{2, i_2, r_1} L_{a_4, i_2}}^T C P_L L_{a_2, i_1} H_{a_1} H_{a_3} \lambda T_{3, a_1, a_2, a_3, a_4}}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} H_{a_1} H_{a_2} S_{1, j_1} y_{1, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} \lambda T_{3, a_1, a_2, j_1}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_2, r_1} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$
									$+\frac{2 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda T_{3, a_3, i_1}} C P_R \overline{L_{a_4, i_2}}^T y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} H_{a_1} H_{a_2} S_{1, j_1} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} \lambda T_{3, a_1, a_2, j_1}^\dagger$	$-\frac{9 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger g_F^2 S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$
$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{\lambda T_{3, a_1, i_1}} S_2 y_{1, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger}{\sqrt{3}}$									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$		
cont.									$-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} \overline{\lambda T_{3, a_1, i_1}} S_2 y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{\sqrt{3} M_F}$	
									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$	$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$	
									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{2, i_1, r_1} \mathcal{I} T_{3, a_1, a_2, j_1} S_2^\dagger}{4 M_F}$	
cont.									$-\frac{9 h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{2, i_2, r_1} y_{2, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} g_F^2 S_2 \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	
									$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{\sqrt{3} M_F}$	
									$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{y_{1, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{1, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{y_{2, i_1, r_2} y_{2, i_3, r_2} y_{2, i_2, r_1} L_{a_2, i_2}}^T C P_L L_{a_1, i_1} S_2 y_{2, i_3, r_2} \mathcal{I} T_{3, a_1, a_2, j_1} S_{1, j_1}^\dagger}{4 M_F}$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$	
cont.									$-\frac{\sqrt{6} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \ell^\dagger g_2^2 S_{1, j_2} y_{1, i_1, r_1} y_{2, i_2, r_1} \overline{I T \tau}_{a_1, a_2, j_2, j_1} S_{2, j_1}^\dagger}{M_F}$		
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{1, i_3, r_2} y_{2, i_1, r_2} y_{2, i_2, r_1} \overline{I T \tau}_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}{2\sqrt{2} M_F}$		
									$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{1, i_1, r_2} y_{1, i_3, r_1} y_{2, i_2, r_2} \overline{I T \tau}_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}{3 M_F}$		
									$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{1, i_1, r_2} y_{2, i_2, r_2} y_{2, i_3, r_1} \overline{I T \tau}_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}{3 M_F}$		
cont.									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{1, i_1, r_2} y_{1, i_2, r_1} y_{2, i_3, r_2} \overline{I T \tau}_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}{2\sqrt{2} M_F}$		
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{1, i_1, r_2} y_{2, i_2, r_1} y_{2, i_3, r_2} \overline{I T \tau}_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}{2\sqrt{2} M_F}$		
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_2, r_2} y_{1, i_3, r_2} y_{2, i_1, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$		
									$+ \frac{2\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger g_2^2 S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_2} \epsilon_{r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{M_F}$		
cont.									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_3, r_2} y_{2, i_1, r_2} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$		
									$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_1} y_{2, i_2, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$		
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{2, i_2, r_2} y_{2, i_3, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{3 M_F}$		
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{1, i_3, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$		
cont.									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_2} y_{2, i_2, r_1} y_{2, i_3, r_2} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$		
									$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$		
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$		
									$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{1, i_1, r_1} y_{2, i_2, r_2} \epsilon_{r_1} \overline{I T \tau}_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$		
cont.									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$		
									$- \frac{3\sqrt{\frac{3}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$		
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$		
									$- \frac{3\sqrt{\frac{3}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{1, i_1, r_1} y_{2, i_2, r_1} \overline{I T \tau}_{3, a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$		
cont.									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$		
									$- \frac{7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$		
									$- \frac{7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$		
									$- \frac{\sqrt{23} M_F}{\sqrt{23} M_F}$		
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\frac{2\sqrt{\frac{3}{8}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{\lambda_{T3}} S_{2, j_2} y_{1, i_1, r_1} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger}{\sqrt{3}}$	$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{\lambda_{T3}} S_{2, j_2} y_{1, i_1, r_1} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j_1}^\dagger}{\sqrt{3}}$	$- \frac{\sqrt{6} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger y_2^2}{M_F}$	
cont.								$\frac{2\sqrt{\frac{3}{8}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3, i_1}} C P_R \overline{L_{a_4, i_2}}^\dagger y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$	$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} H_{a_1} H_{a_2} S_{1, j_2} y_{2, i_1, r_1} \overline{I T \tau}_{3, a_1, a_2, j_2, j_1} \lambda_{T3} S_{2, j_1}^\dagger}{\sqrt{3}}$	$- \frac{2\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_1} \epsilon_{a_1, a_2} S_{1, j_1}^\dagger y_2^2}{M_F}$	
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} H_{a_1} H_{a_2} S_{1, j_2} y_{1, i_1, r_1} \overline{I T \tau}_{3, a_1, a_2, j_2, j_1} \lambda_{T3} S_{2, j_1}^\dagger}{\sqrt{3}}$	$- \frac{\sqrt{6} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{1, i_1, r_1} y_{2, i_2, r_1} \overline{I T \tau}_{3, a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger y_2^2}{M_F}$	
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} H_{a_1} H_{a_2} S_{1, j_2} y_{2, i_1, r_1} \overline{I T \tau}_{3, a_1, a_2, j_2, j_1} \lambda_{T3} S_{2, j_1}^\dagger}{\sqrt{3}}$	$- \frac{2\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{1, i_1, r_1} y_{2, i_2, r_1} \epsilon_{a_1, a_2} S_{2, j_1}^\dagger y_2^2}{M_F}$	
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} H_{a_1} H_{a_2} S_{1, j_2} y_{2, i_1, r_1} \overline{I T \tau}_{3, a_1, a_2, j_2, j_1} \lambda_{T3} S_{2, j_1}^\dagger}{\sqrt{3}}$	$+ \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} y_2^2 S_{2, j_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2 M_F}$	
cont.									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{1, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_2} \overline{I T \tau}_{a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2} M_F}$	$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{1, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{1, i_3, r_2} \overline{I T \tau}_{a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3 M_F}$	
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} y_{2, i_1, r_2} y_{2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2} M_F}$	$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} y_{2, i_1, r_2} y_{2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3 M_F}$	
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{1, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2} M_F}$	$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{1, i_2, r_2} y_{2, i_3, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3 M_F}$	
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2} M_F}$	$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3 M_F}$	
cont.									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2} M_F}$	$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3 M_F}$	
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2} M_F}$	$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3 M_F}$	
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2} M_F}$	$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3 M_F}$	
									$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{2\sqrt{2} M_F}$	$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1} y_{2, i_3, r_2} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} y_{2, i_3, r_2} \overline{I T \tau}_{3, a_1, a_2, j_1, j_2} S_{1, j_1}^\dagger}{3 M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	S_1S_2FH	$S_1S_2F\ell$
cont.										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4M_F}$
cont.										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} C P_R \overline{L_{a_3,i_3}} S_{1,j_2} y_{1,i_3,r_2} y_{1,i_1,r_2} y_{2,i_1,r_2} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
cont.										$+ \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} g_{\gamma}^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{2,j_1}^\dagger}{2M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
										$\frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{2,j_1}^\dagger}{3M_F}$
cont.										$\frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{2,j_1}^\dagger}{3M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{2,j_1}^\dagger}{2\sqrt{2}M_F}$
cont.										$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} g_{\gamma}^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{3M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{3M_F}$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.										$+ \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_2} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$
										$+ \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
cont.										$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$
										$- \frac{3\sqrt{\frac{2}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_2} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$
cont.										$\frac{7h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$
										$- \frac{7h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$
										$- \frac{\sqrt{6} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} g_{\gamma}^2 S_{2,j_2} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger}{M_F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$			$- \frac{2\sqrt{\frac{2}{3}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}^{-r} C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda T \overline{a_{a_1,a_2} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}}{M_F}$
										$- \frac{2\sqrt{\frac{2}{3}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda T \overline{a_{a_3,a_1}}} C P_R \overline{L_{a_4,i_2}} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} I T \overline{a_{a_1,a_2,j_2,j_1}} \lambda T \overline{a_{a_3}} S_{2,j_1}^\dagger}{\sqrt{3}}$
										$- \frac{\sqrt{6} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} g_{\gamma}^2 S_{2,j_2} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger}{2\sqrt{2}M_F}$
										$- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}^{-r} C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} I T \overline{a_{a_1,a_2,j_2,j_1}} S_{1,j_1}^\dagger}{3M_F}$

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	S_1S_2FH	$S_1S_2F\ell$
cont.									$\frac{3\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}}^{\dagger}\lambda_{12}^{(2)}}{M_F}$	$\frac{3\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}}^{\dagger}\lambda_{12}^{(2)}}}{M_F}$
									$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^{\dagger}\lambda_{12}^{(2)}}}{\sqrt{11}M_F}$
cont.									$\frac{7h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^{\dagger}\lambda_{12}^{(3)}}{\sqrt{23}M_F}$	$\frac{7h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^{\dagger}\lambda_{12}^{(3)}}}{\sqrt{23}M_F}$
									$-\frac{7h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^{\dagger}\lambda_{12}^{(3)}}}{\sqrt{23}M_F}$	$-\frac{7h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^{\dagger}\lambda_{12}^{(3)}}}{\sqrt{23}M_F}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_{T3}S_{2,j_2}y_{1,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1,j_2}H_{a_1}^{\dagger}H_{a_2}^{\dagger}S_{1,j_1}^{\dagger}}}}{\sqrt{3}}$	$\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{M_F}$
								$\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}CPH\overline{L_{a_4,i_2}}^{\dagger}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^{\dagger}H_{a_2}^{\dagger}}}{M_F}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_{T3}S_{2,j_2}y_{2,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1,j_2}H_{a_1}^{\dagger}H_{a_2}^{\dagger}S_{1,j_1}^{\dagger}}}}{\sqrt{3}}$	$\frac{2\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}g_1^2S_{2,j_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{M_F}$
cont.								$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{2\sqrt{2}M_F}$
								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{3M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{3M_F}$
cont.								$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{2\sqrt{2}M_F}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{2\sqrt{2}M_F}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{2\sqrt{2}M_F}$
								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{3M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{3M_F}$
cont.								$\frac{2\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}g_1^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^{\dagger}}{M_F}$	$\frac{2\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^{\dagger}}{M_F}$	$\frac{2\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}g_1^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^{\dagger}}{M_F}$
								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{4M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{4M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{4M_F}$
cont.								$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{4M_F}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{4M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{4M_F}$
								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{4M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{4M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{4M_F}$
cont.								$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^{\dagger}}}{3M_F}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{2\sqrt{2}M_F}$	$\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}g_2^2S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{M_F}$
								$-\frac{2\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{IT_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}g_1^2S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{M_F}$	$-\frac{2\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{IT_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}g_1^2S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{M_F}$	$-\frac{2\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{IT_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}g_1^2S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{M_F}$
cont.								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{2\sqrt{2}M_F}$	$\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$
								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{2\sqrt{2}M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{2\sqrt{2}M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{2\sqrt{2}M_F}$
cont.								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$
								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$
cont.								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$
								$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^{\dagger}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^{\dagger}}}}{3M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ
										$S_1 S_2 F$
										$S_1 S_2 F \ell$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$+ \frac{\sqrt{\frac{3}{2}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_3,i_1} S_{2,j_2} \mathcal{I} T 3_{a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$
cont.										$+ \frac{\sqrt{\frac{3}{2}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T 3_{a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$
										$3 \sqrt{\frac{2}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I} T 3_{a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}$
cont.										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$
										$3 \sqrt{\frac{2}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T 3_{a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$
cont.										$7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(3)}$
										$- \frac{7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$

Field combinations 10 of 10

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell e$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	e
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			